Diffusion Seminar (Week3)

Generative Modeling II

In-depth VAE and Introduction to DDPM

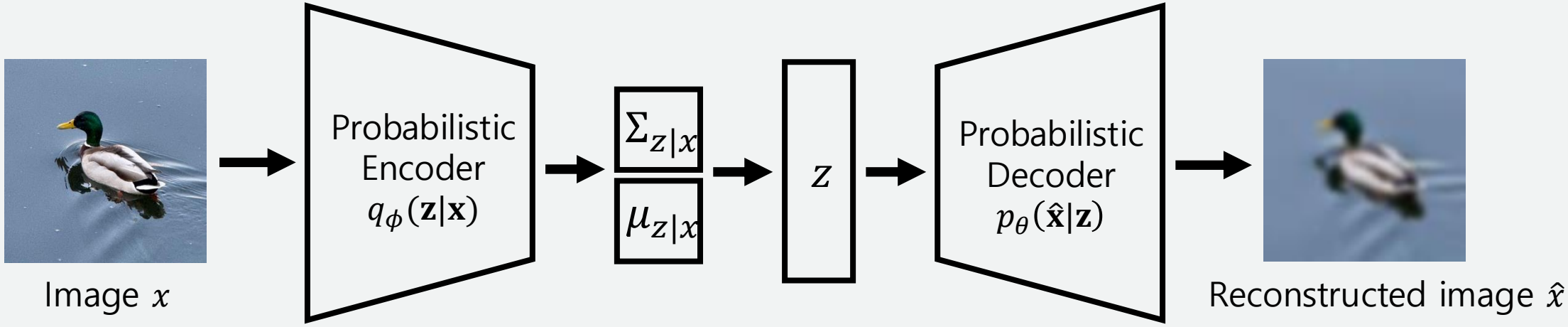
Jaeik Kim

SNU AIDAS Lab

AIDAS Lab

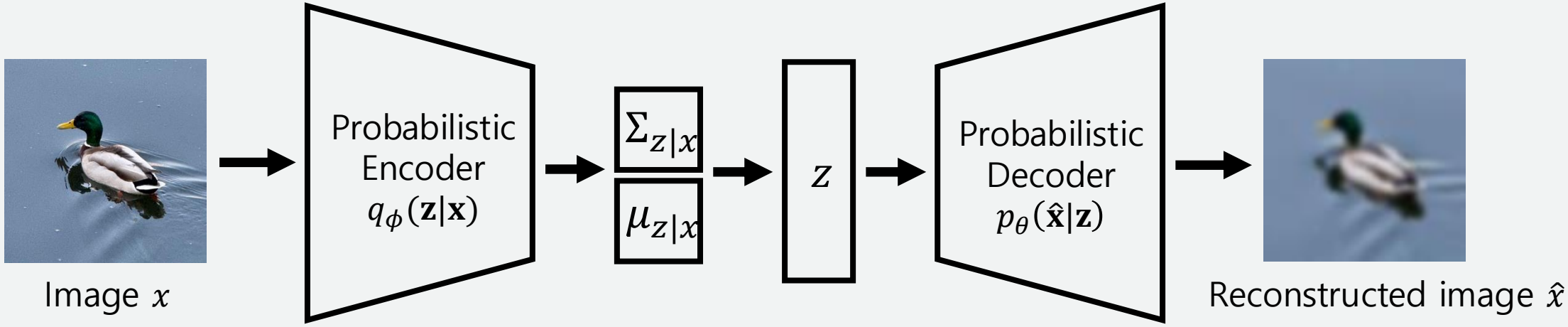
Recap: VAE

Previous Seminar - VAE



Previous Seminar - VAE

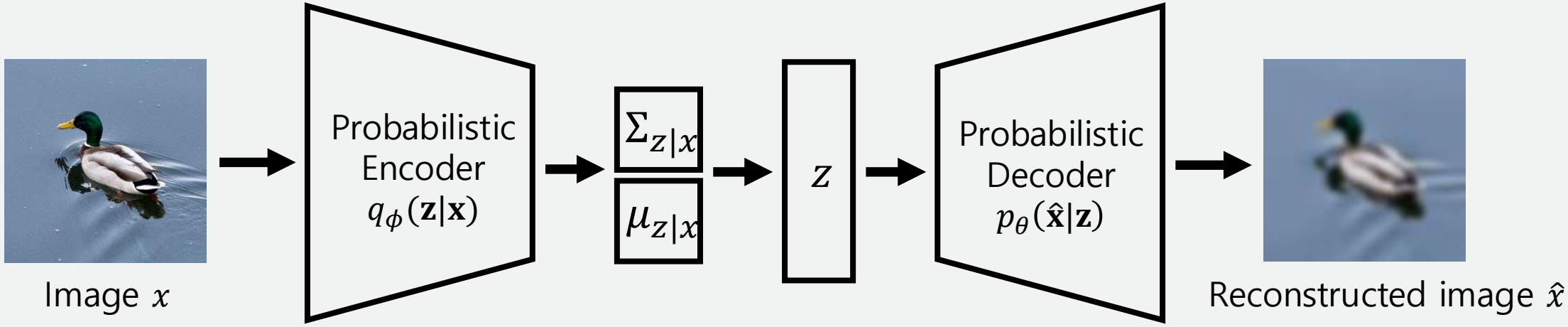
Objective: $\log p_\theta(x) \geq \mathbb{E}_z[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x)||p(z)\right)$



Previous Seminar - VAE

Reconstruction Term

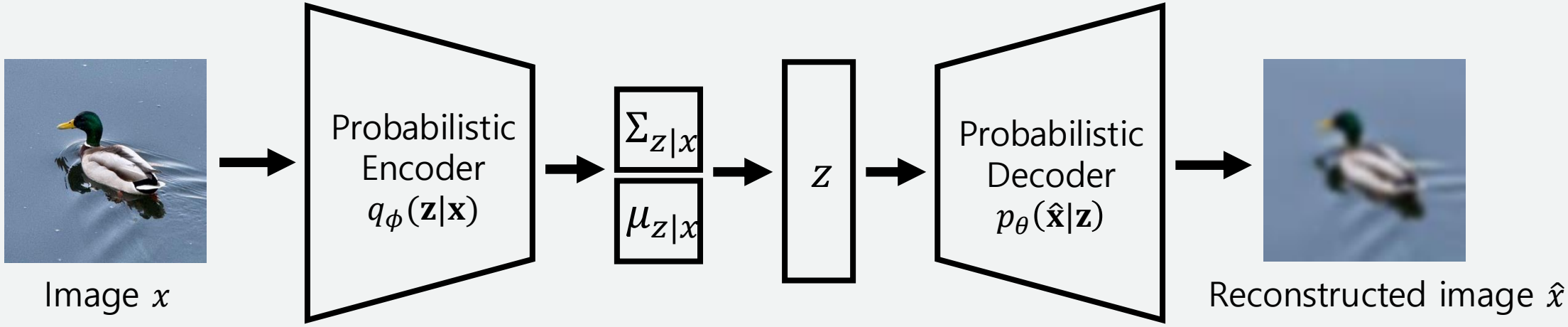
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Previous Seminar - VAE

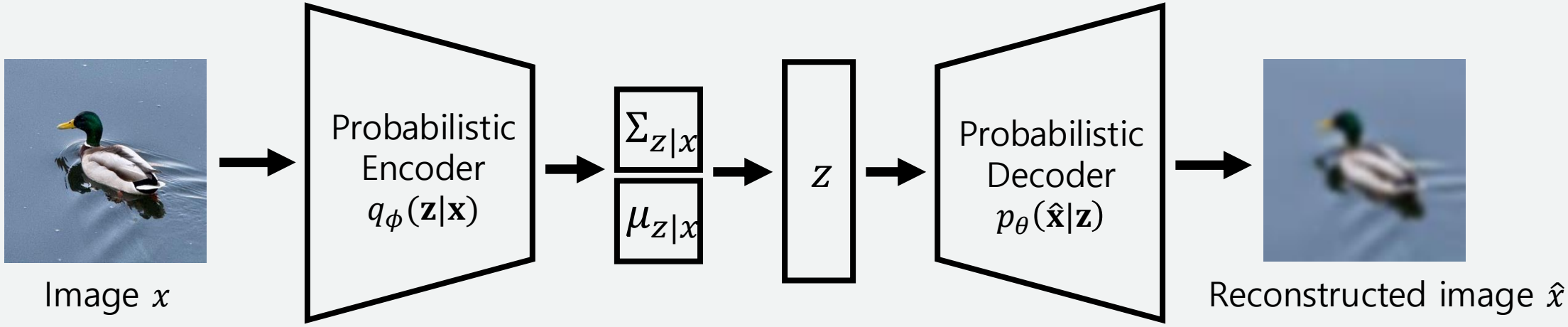
Reconstruction Term Prior Fitting Term = Regularization Term

Objective: $\log p_\theta(x) \geq \mathbb{E}_z[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x)||p(z)\right)$



Previous Seminar - VAE

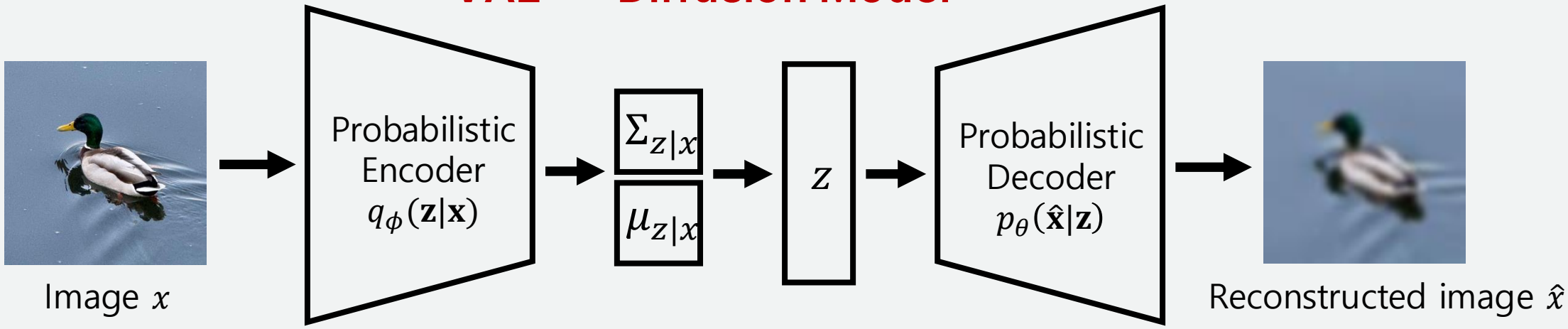
How can we optimize VAE?



Previous Seminar - VAE

How can we optimize VAE?

VAE ↔ Diffusion Model



Today

1. Basics: Likelihood, MLE, MAP, Variational Inference

2. Deep dive into VAE: Optimization and Generation

3. DDPM(Denoising Diffusion Probabilistic Model) Overview

Basics

Likelihood

Likelihood measures how well a statistical model explains observed data by calculating the probability of seeing that data under different(i.e., unknown) parameter values of the model.

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= How likely it is to generate the observed data?

Likelihood vs. Probability

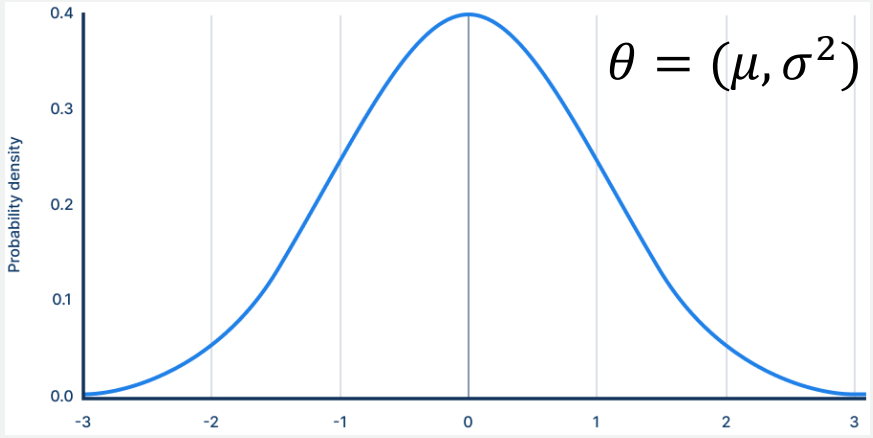
Likelihood vs. Probability

Likelihood

Probability

Likelihood vs. Probability

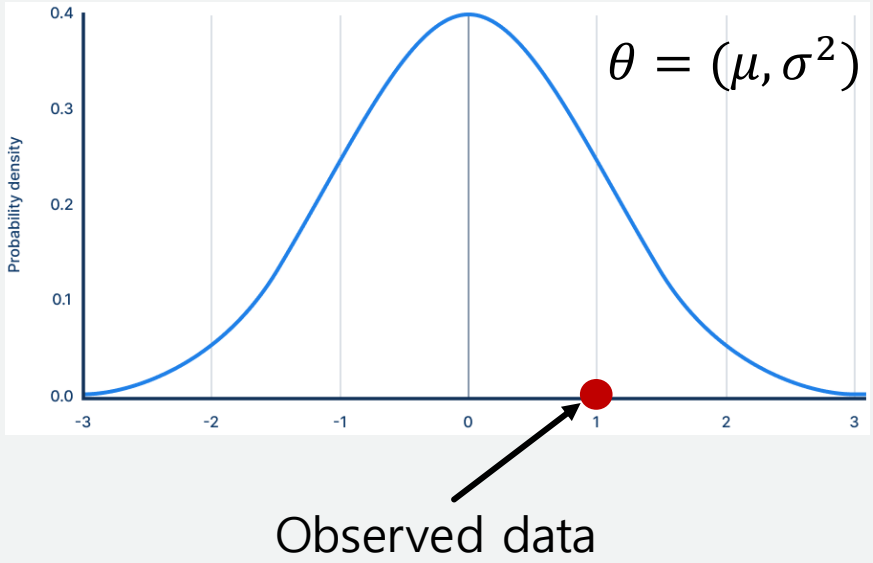
Likelihood



Probability

Likelihood vs. Probability

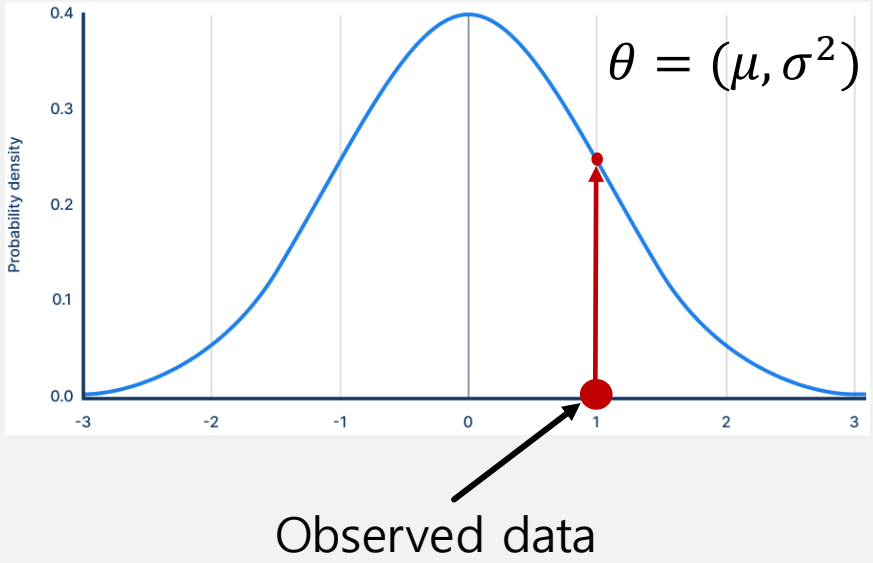
Likelihood



Probability

Likelihood vs. Probability

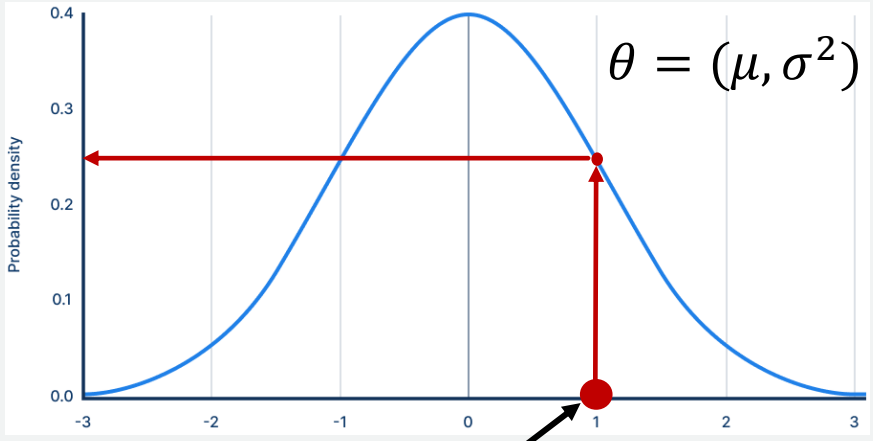
Likelihood



Probability

Likelihood vs. Probability

Likelihood



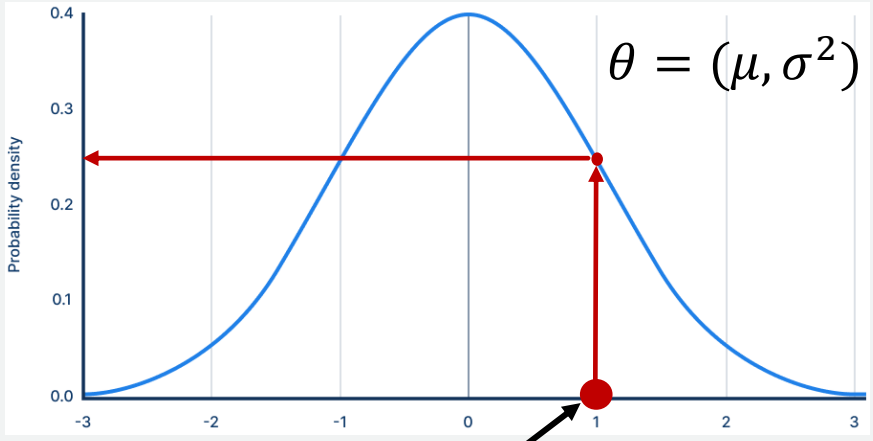
Observed data

$P(x|\theta) = 0.24$

Probability

Likelihood vs. Probability

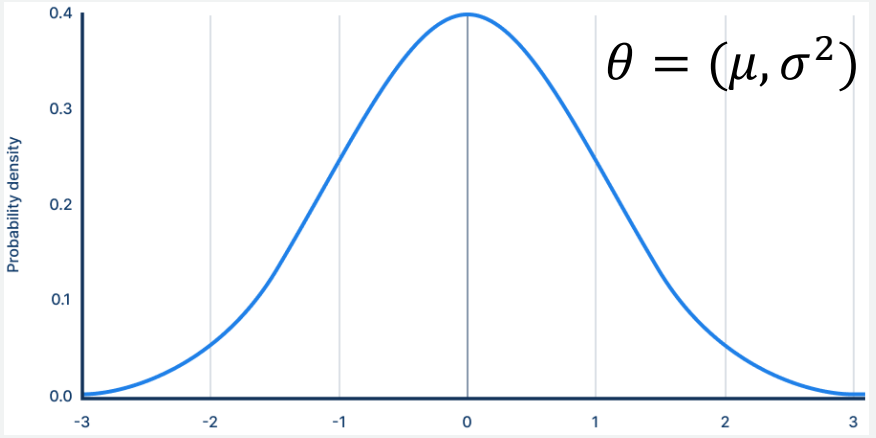
Likelihood



Observed data

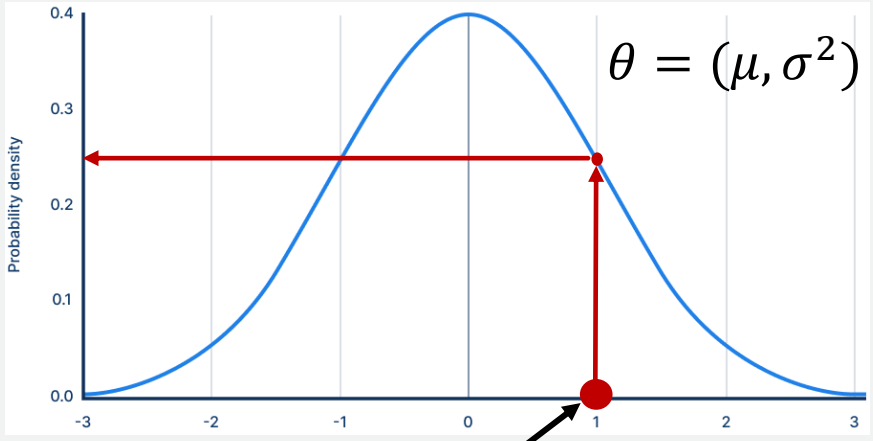
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Likelihood vs. Probability

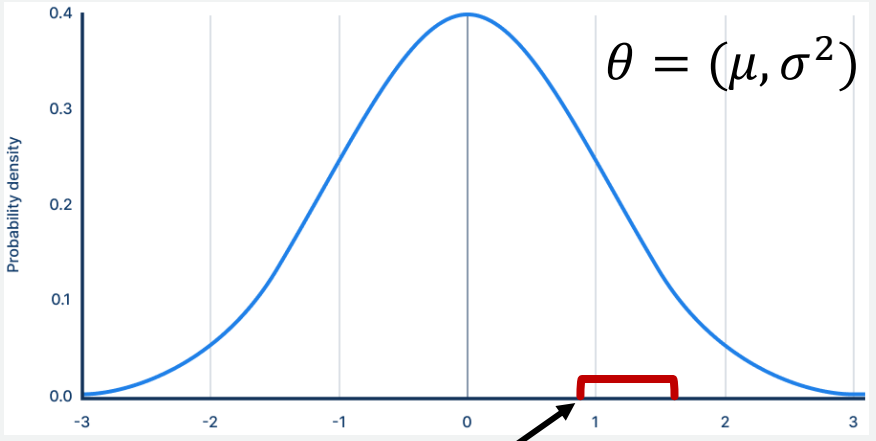
Likelihood



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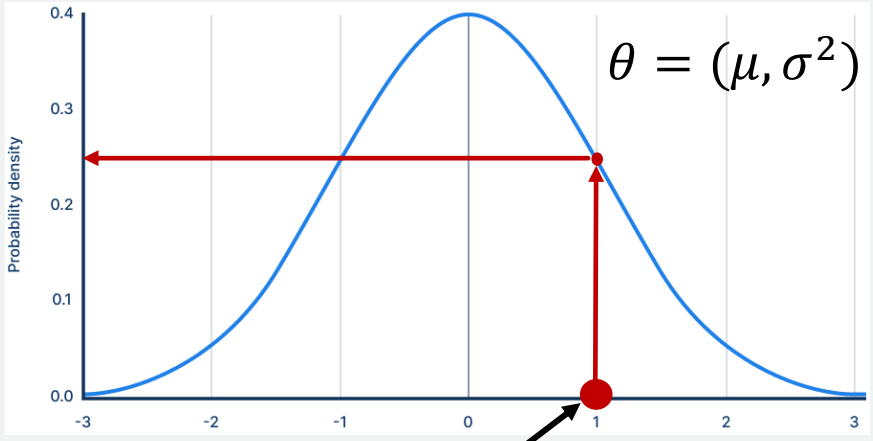
Probability



Expected event

Likelihood vs. Probability

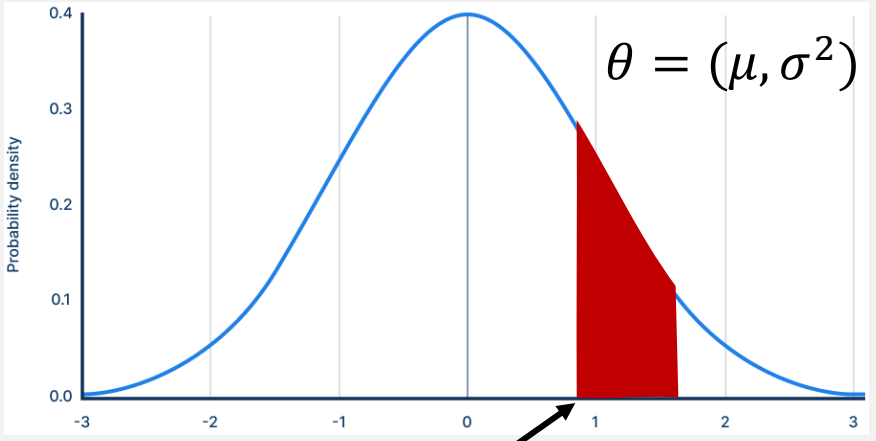
Likelihood



Observed data

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Probability



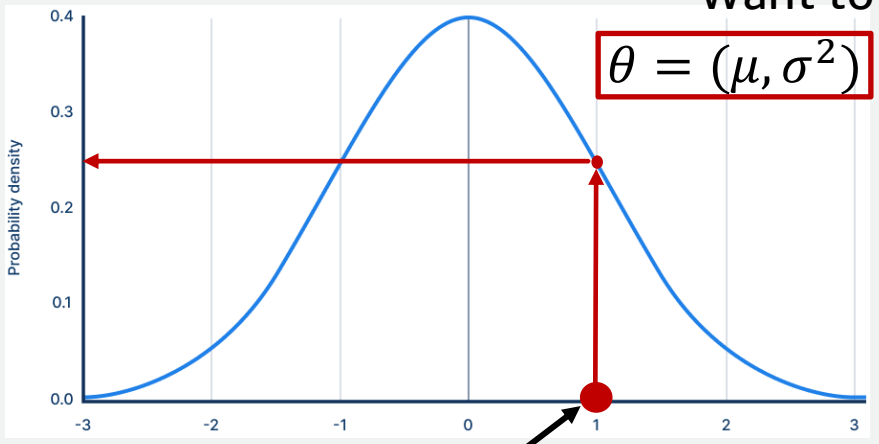
Expected event

$P(x|\theta) = 0.21$

Likelihood vs. Probability

Likelihood

Want to find

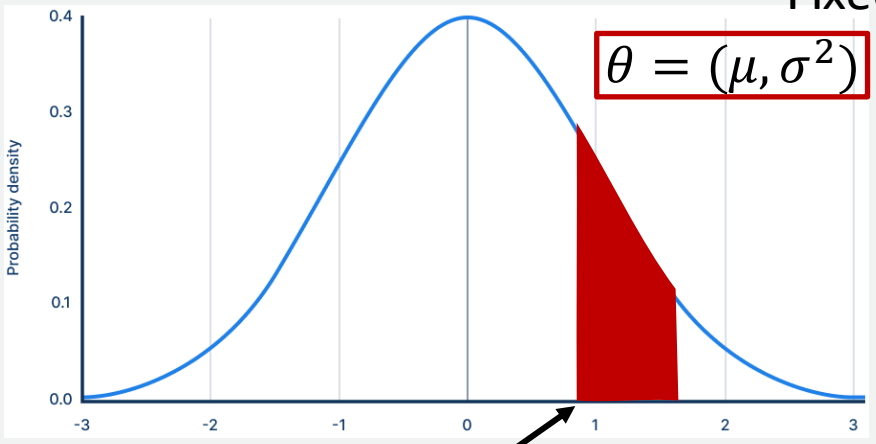


Observed data

$P(x|\theta) = 0.24$

Probability

Fixed



Expected event

$P(x|\theta) = 0.21$

Maximum Likelihood Estimation(MLE)

MLE aims to estimate the parameters of a probability distribution by finding the parameter values that maximize the likelihood of the observed data.

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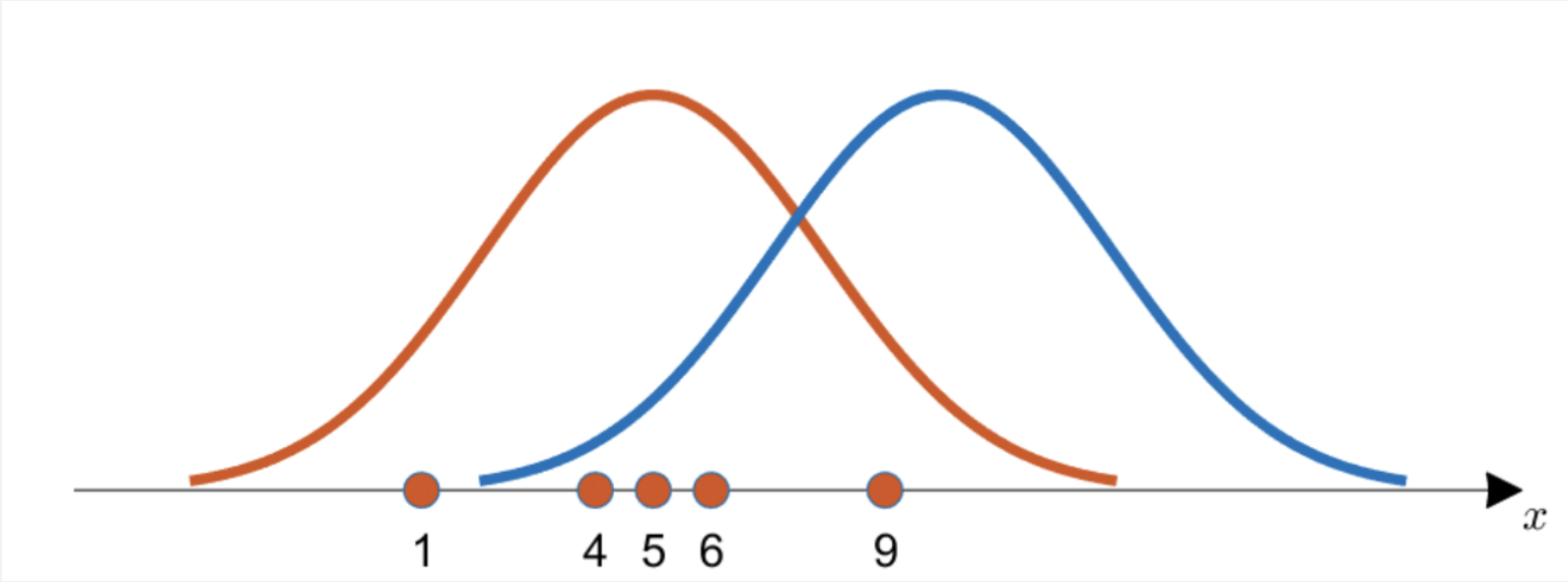
All generative models aims to maximize likelihood of observed data, either implicit or explicit ways.

$$\underset{\theta}{\operatorname{argmax}}\, P(x|\theta) \rightarrow P_\theta(x) \approx P_{data}(x)$$

Maximum Likelihood Estimation(MLE)

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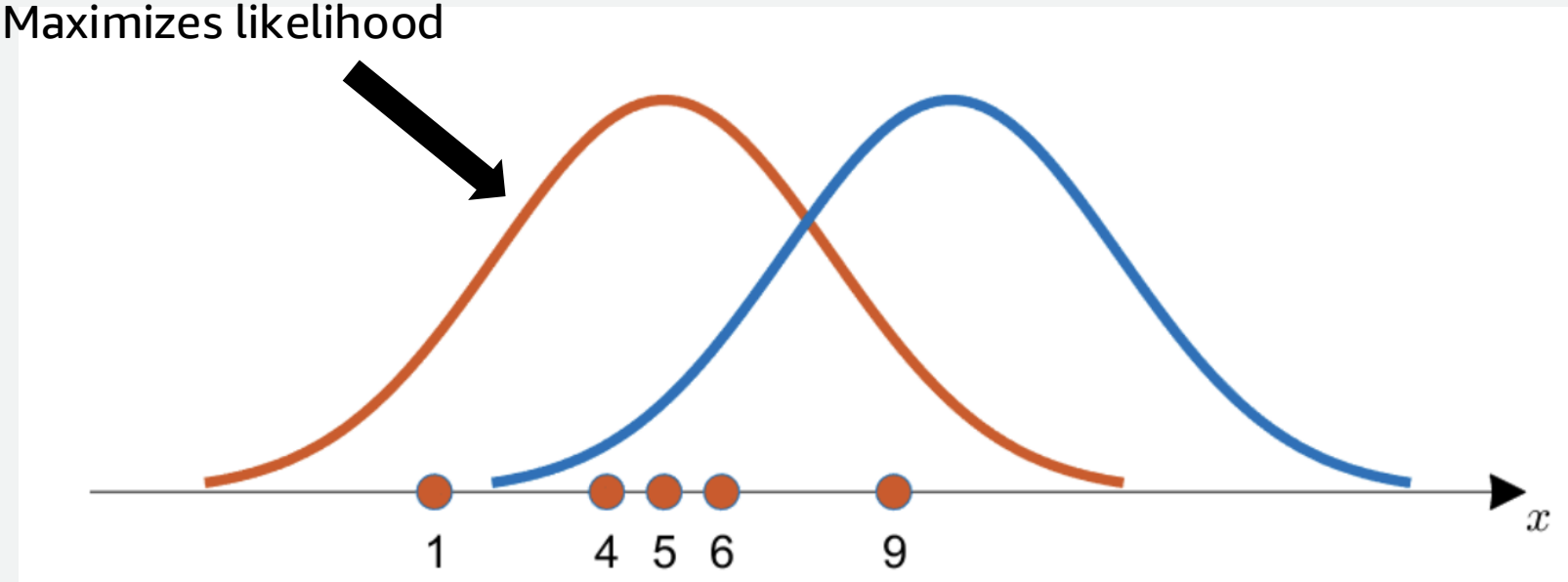
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MLE:

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Overfitting to observed data

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Overfitting to observed data

Example: Coin flip

Head	Tail	$\boldsymbol{\theta}^*$
10	0	1.0
8	2	0.8

Maximum Likelihood Estimation(MLE)

MLE:

$$\underset{\theta}{\operatorname{argmax}} P(x|\theta) \rightarrow P_\theta(x) \approx P_{data}(x)$$

Overfitting to observed data

Example: Coin flip

Head	Tail	θ^*
10	0	1.0
8	2	0.8
8	2	0.65

More reasonable!

Maximum A Posterior(MAP)

MAP: Maximize posterior $\mathrm{P}(\theta|\mathrm{x})$

- What is the likely θ given data x?
- Evaluates how likely θ is to be correct, given the observed data.

Maximum A Posterior(MAP)

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$$\underset{\theta}{\operatorname{argmax}}\, P(\theta|x)$$

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$$\underset{\theta}{\operatorname{argmax}}\, P(\theta|x) = \underset{\theta}{\operatorname{argmax}} \frac{P(x|\theta)P(\theta)}{P(x)}$$

By Bayesian Rule

Maximum A Posterior(MAP)

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$$\underset{\theta}{\operatorname{argmax}}\, P(\theta|x) = \underset{\theta}{\operatorname{argmax}} \frac{P(x|\theta)P(\theta)}{P(x)} \propto \underset{\theta}{\operatorname{argmax}}\, P(x|\theta)P(\theta)$$

Constant

Maximum A Posterior(MAP)

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$$\underset{\theta}{\operatorname{argmax}} \underbrace{P(\theta|x)}_{\text{Posterior}} = \underset{\theta}{\operatorname{argmax}} \frac{\overbrace{P(x|\theta)}^{\text{Likelihood}}\overbrace{P(\theta)}^{\text{Prior}}}{\underbrace{P(x)}_{\text{Evidence}}} \propto \underset{\theta}{\operatorname{argmax}} P(x|\theta)P(\theta)$$

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Example: Coin flip

Head	Tail	MLE	Prior	MAP
10	0	1.0	0.5	0.75
8	2	0.8	0.5	0.65

Relationship between MLE and MAP

MLE: Maximize likelihood $\mathrm{P}(\mathrm{x}|\theta)$

$$\underset{\theta}{\operatorname{argmax}}\, P(x|\theta)$$

MAP: Maximize posterior $\mathrm{P}(\theta|\mathrm{x})$

$$\underset{\theta}{\operatorname{argmax}}\, P(x|\theta)\underline{P(\theta)}$$

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If $\mathrm{P}(\theta)$ is uniform(*i.e.,* constant), then MAP = MLE

MLE and MAP

	MLE	MAP
Objective	$\underset{\theta}{\operatorname{argmax}}\, P(x\|\theta)$	$\underset{\theta}{\operatorname{argmax}}\, P(\theta\|x)$
Prior	Not considerd	Considered
Overfitting	Highly	Reduced
Estimate	Point	Point
Uncertainty	Not considerd	Not considerd

MLE and MAP

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Why it is important?

MLE and MAP

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Why it is important?

1. Reliability
2. Sample diversity(generative model)

Variational Inference(VI)

MAP: Maximize posterior $\mathrm{P}(\theta|\mathrm{x})$

$$\underset{\theta}{\operatorname{argmax}}\, P(\theta|x) = \underset{\theta}{\operatorname{argmax}} \frac{P(x|\theta)P(\theta)}{P(x)} \propto \underset{\theta}{\operatorname{argmax}}\, P(x|\theta)P(\theta)$$

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VI seeks to approximate the full posterior, rather than maximizing a point.

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$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)} = \frac{P(x|\theta)P(\theta)}{\int P(x|\theta)P(\theta)d\theta}$$

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$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)} = \frac{P(x|\theta)P(\theta)}{\int P(x|\theta)P(\theta)d\theta}$$ Evidence

Variational Inference(VI)

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Intractable!

Variational Inference(VI)

$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)} = \frac{P(x|\theta)P(\theta)}{\int P(x|\theta)P(\theta)d\theta}$$ ⟵ Bayesian parameter inference

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$$P(z|x) = \frac{P(x|z)P(z)}{P(x)} = \frac{P(x|z)P(z)}{\int P(x|z)P(z)dz}$$ ⟵ Latent variable inference

VAE, Diffusion

Variational Inference(VI)

$Q(z|\lambda) \approx P(z|x)$ ⟵ Let's approximate $P(z \mid x)$ with tractable Q

Variational Inference(VI)

$$Q(z|\lambda) \approx P(z|x) \leftarrow \min_{\lambda} D_{KL}(Q(z|\lambda) \,||\, P(z|x))$$

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$$\geq \boxed{\mathbb{E}_Q[\log P(x,z)] - \mathbb{E}_Q[log\ Q(z|\lambda)]} \quad \text{ELBO}$$

Variational Inference(VI)

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$$\therefore \log P(x) = D_{KL}(Q(z|\lambda) \,||\, P(z|x)) + \mathbb{E}_Q[\log P(x,z)] - \mathbb{E}_Q[log\ Q(z|\lambda)]$$

$$\geq \mathbb{E}_Q[\log P(x,z)] - \mathbb{E}_Q[log\ Q(z|\lambda)]$$

$$= \mathbb{E}_Q[\log P(x|z)] + \mathbb{E}_Q[\log P(z)] - \mathbb{E}_Q[log\ Q(z|\lambda)] \quad \cdots \text{ (By Bayesian)}$$

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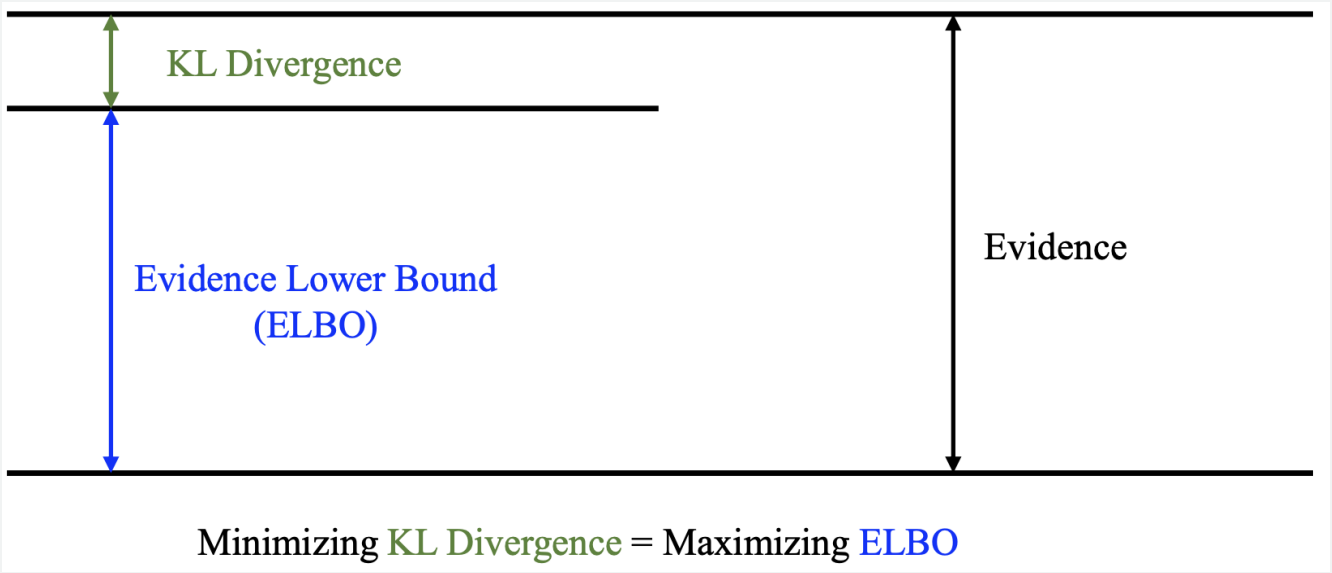
$$= \mathbb{E}_Q[\log P(x|z)] + \mathbb{E}_Q[\log P(z)] - \mathbb{E}_Q[log\ Q(z|\lambda)] \quad \cdots \text{ (By Bayesian)}$$

$$= \mathbb{E}_Q[\log P(x|z)] - D_{KL}(Q(z|\lambda)||P(z)) \quad \cdots \text{ (By definition)}$$

Variational Inference(VI)

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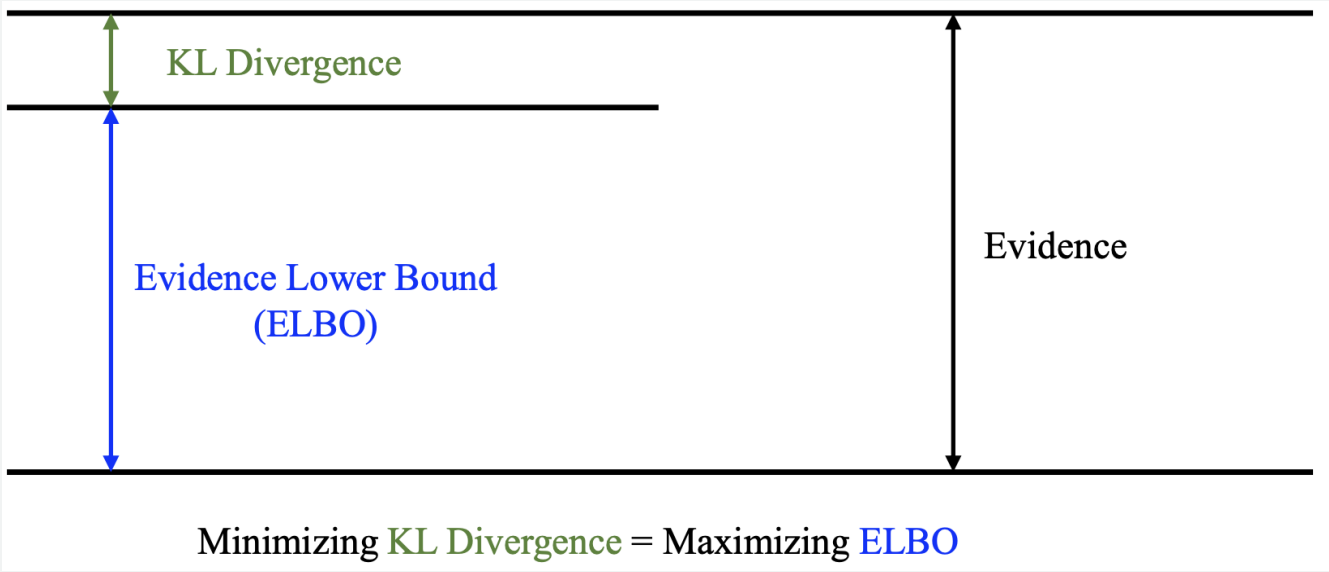


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Resource:
IWAE
Normalizing flow



Amortized Variational Inference(AVI)

$$\log P(x) \geq \mathbb{E}_Q[\log P(x|z)] - D_{KL}(Q(z|\lambda)||P(z))$$

Amortized Variational Inference(AVI)

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Amortized Variational Inference(AVI)

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We have to optimize each λ_i given every x_i

Amortized Variational Inference(AVI)

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We have to optimize each λ_i given every x_i

Let's model ϕ that satisfies $f_\phi(x) = \lambda_i$

Deep dive into VAE

Variational Autoencoder(VAE)

VI: $\log P(x) = D_{KL}(Q(z|\lambda) \,||\, P(z|x)) + \mathbb{E}_Q[\log P(x|z)] - D_{KL}(Q(z|\lambda)||P(z))$

Variational Autoencoder(VAE)

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$P(x) \rightarrow p_\theta(x)$ (Decoder)

$Q(z|\lambda) \rightarrow q_\phi(z|x)$ (Encoder w/ amortized)

Variational Autoencoder(VAE)

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VAE: $\log p_\theta(x) = D_{KL}\left(q_\phi(z|x) \,||\, p_\theta(z|x)\right) + \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x)||p(z)\right)$

Variational Autoencoder(VAE)

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$\geq \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - KL\left(q_\phi(z|x)||p(z)\right)$

Variational Autoencoder(VAE)

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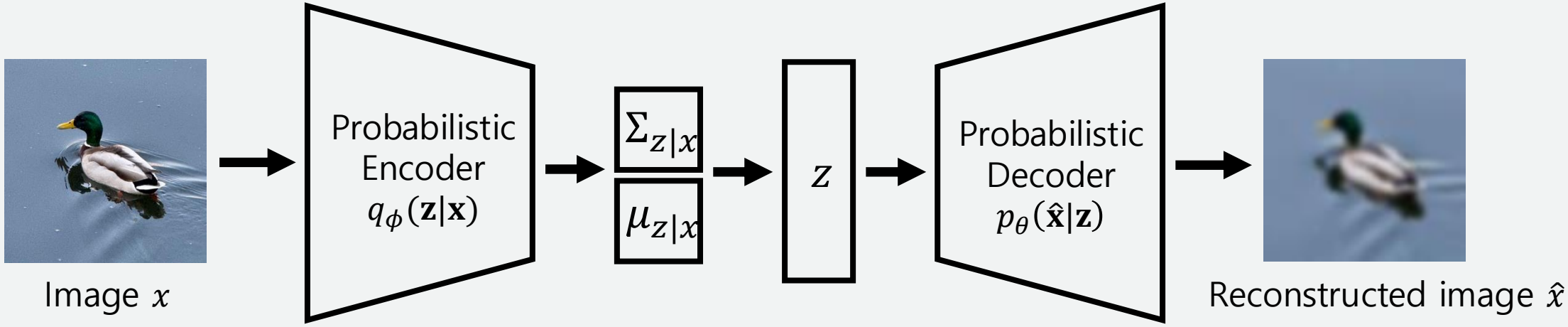
$Q(z|\lambda) \rightarrow q_\phi(z|x)$ (Encoder w/ amortized)

VAE: $\log p_\theta(x) = D_{KL}\left(q_\phi(z|x) \,||\, p_\theta(z|x)\right) + \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x)||p(z)\right)$

$$\geq \boxed{\mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - KL\left(q_\phi(z|x)||p(z)\right)}$$

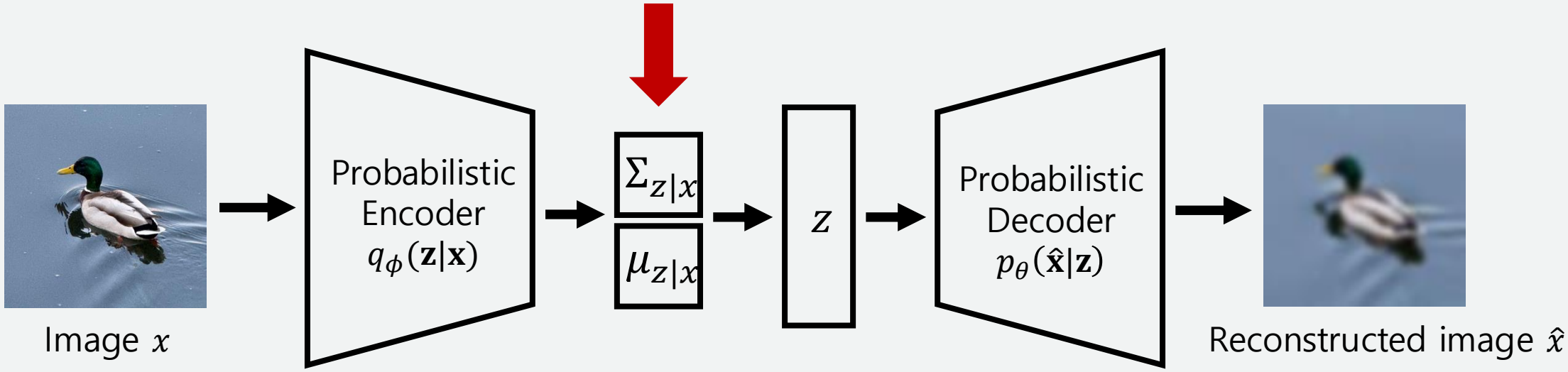
How can we maximize it?

VAE Optimization



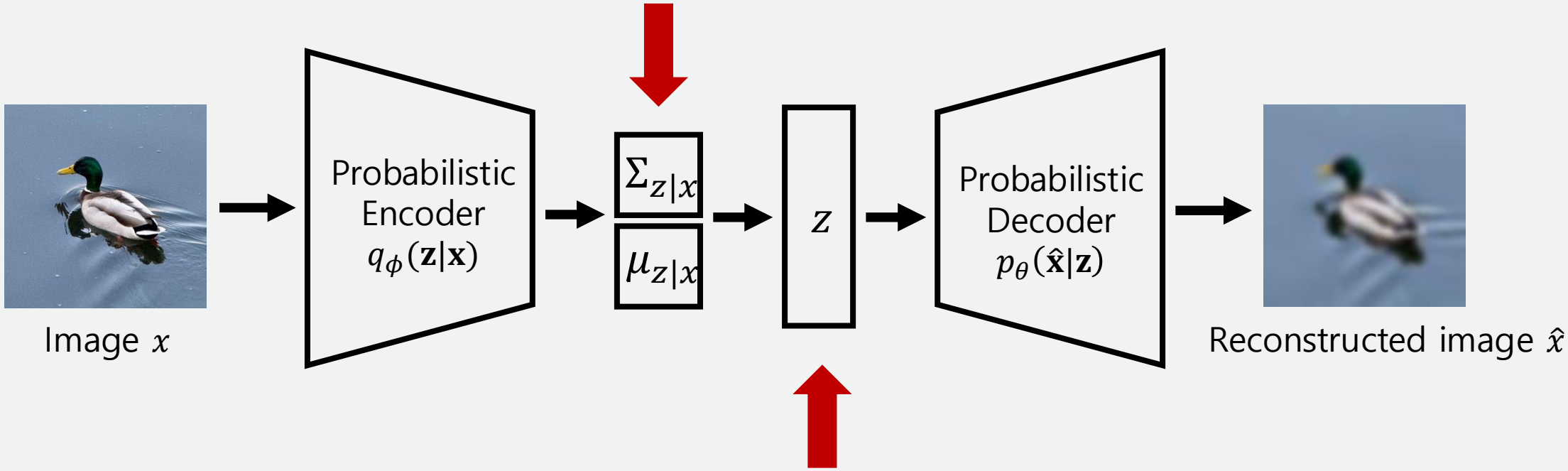
VAE Optimization

Assumption1: $q_\phi(\mathbf{z}|\mathbf{x})$ follows multivariative Gaussian distribution



VAE Optimization

Assumption1: $q_\phi(\mathbf{z}|\mathbf{x})$ follows multivariative Gaussian distribution



Assumption2: $\mathrm{p}(z)$ follows multivariative normal Gaussian distribution

VAE Optimization – Reconstruction Term

$$ELBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

VAE Optimization – Reconstruction Term

$$\mathrm{E}LBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

$$\mathbb{E}_{q_\phi}[\log p_\theta(x|z)]$$

VAE Optimization – Reconstruction Term

$$\mathrm{E}LBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

$$\mathbb{E}_{q_\phi}[\log p_\theta(x|z)] = \int \log p_\theta(x|z)\, q_\phi(z|x)\, dz$$

VAE Optimization – Reconstruction Term

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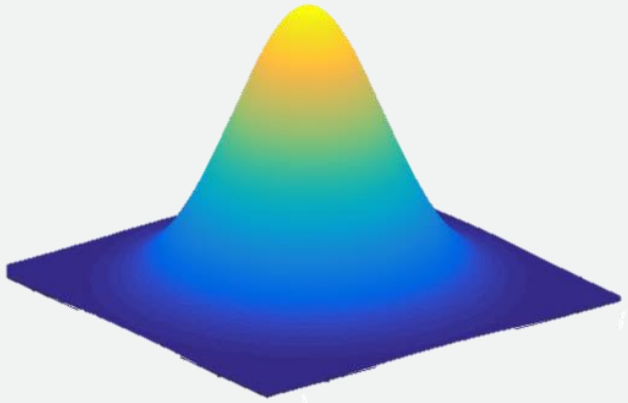
$$\mathbb{E}_{q_\phi}[\log p_\theta(x|z)] = \int \log p_\theta(x|z)\, q_\phi(z|x)\, dz \approx \frac{1}{L}\sum_{l=1}^{L} \log p_\theta\left(x\middle|z^{(l)}\right) \quad \cdots \text{ (By Monte-Carlo tech.)}$$

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[Monte-Carlo Tecnique(Estimation)]



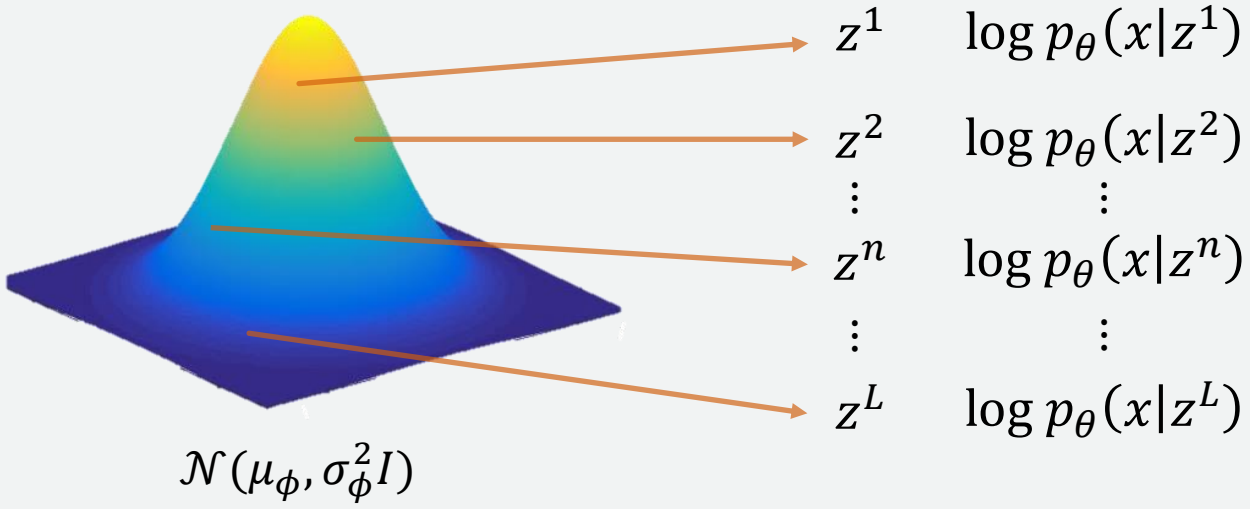
$\mathcal{N}(\mu_\phi, \sigma_\phi^2 I)$

VAE Optimization – Reconstruction Term

$$ELBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

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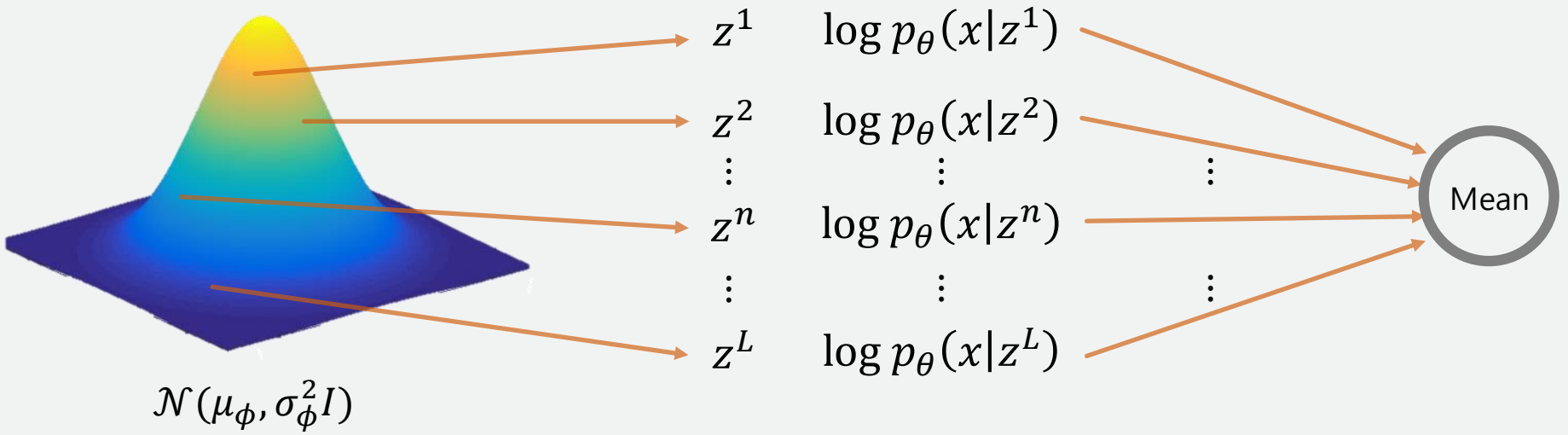


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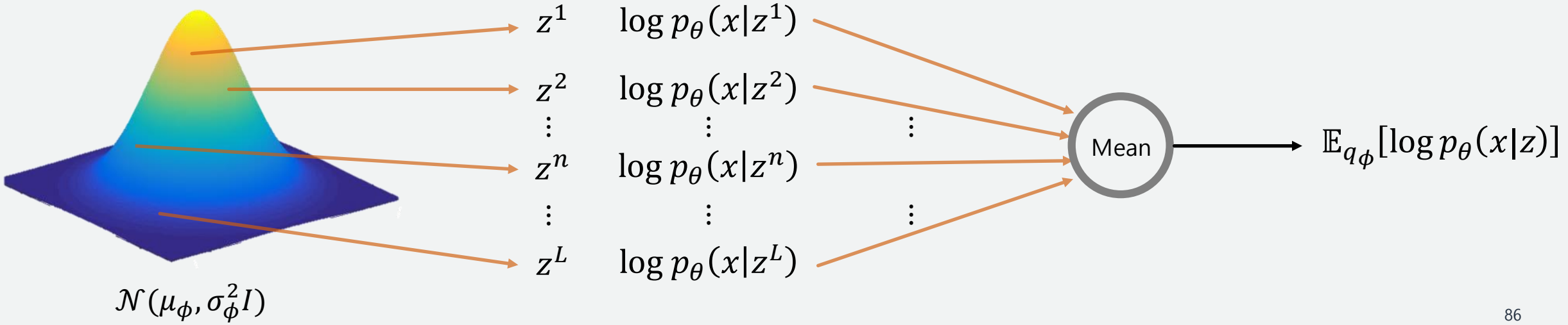


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VAE Optimization – Reconstruction Term

$$\mathrm{E}LBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

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(case1) $p_\theta(x|z)$: *Gaussian*

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$$p_\theta(x|z) \sim \mathcal{N}\left(\mu_\theta, \sigma_\theta^2 I\right)$$

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VAE Optimization – Reconstruction Term

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$$= -\frac{1}{2\sigma^2}||x-\mu_\theta||^2 - \log\sqrt{\left(2\pi\sigma_\theta^2\right)^d}$$

VAE Optimization – Reconstruction Term

$$ELBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

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$$= -\frac{1}{2\sigma^2}||x-\mu_\theta||^2 - \boxed{\log\sqrt{\left(2\pi\sigma_\theta^2\right)^d}}$$

Constant if identity covariance

VAE Optimization – Reconstruction Term

$$ELBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

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$$= \boxed{-\frac{1}{2\sigma^2}||x-\mu_\theta||^2} - \log\sqrt{\left(2\pi\sigma_\theta^2\right)^d}$$

This is why we call it "Reconstruction" term

VAE Optimization – Reconstruction Term

$$\mathrm{E}LBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

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VAE Optimization – Reconstruction Term

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$p_\theta(x|z) \sim Benoulli(p)$

VAE Optimization – Reconstruction Term

$$ELBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

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$$p_\theta(x|z) \sim Benoulli(p)$$

$$\log p_\theta\left(x|z^{(l)}\right) = \log \prod_{j=1}^{D} p_\theta\left(x_j|z^{(l)}\right)$$

VAE Optimization – Reconstruction Term

$$\mathrm{E}LBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

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$$= \sum_{j=1}^{D} \log p_j^{x_j}\left(1-p_j\right)^{1-x_j}$$

VAE Optimization – Reconstruction Term

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$$= \sum_{j=1}^{D} \log p_j^{x_j}\left(1-p_j\right)^{1-x_j} = \sum_{j=1}^{D} x_j \log p_j + \left(1-x_j\right)\log(1-p_j)$$

VAE Optimization – Reconstruction Term

$$ELBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

$$\mathbb{E}_{q_\phi}[\log p_\theta(x|z)] = \int \log p_\theta(x|z)\, q_\phi(z|x)\, dz \approx \frac{1}{L}\sum_{l=1}^{L} \log p_\theta\left(x\middle|z^{(l)}\right) \quad \cdots \text{ (By Monte-Carlo tech.)}$$

(case2) $p_\theta(x|z)$: *Bernoulli*

$$p_\theta(x|z) \sim Benoulli(p)$$

$$\log p_\theta\left(x\middle|z^{(l)}\right) = \log \prod_{j=1}^{D} p_\theta\left(x_j\middle|z^{(l)}\right) = \sum_{j=1}^{D} \log p_\theta\left(x_j\middle|z^{(l)}\right)$$

$$= \sum_{j=1}^{D} \log p_j^{x_j}\left(1-p_j\right)^{1-x_j} = \sum_{j=1}^{D} \boxed{x_j \log p_j + \left(1-x_j\right)\log(1-p_j)}$$

-Cross Entorpy

VAE Optimization – Regularization Term

$$\mathrm{E}LBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

VAE Optimization – Regularization Term

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KLD for multivariate gaussian distributions

$$D_{KL}(\mathcal{N}_1 \,||\, \mathcal{N}_2) = \frac{1}{2}(tr(\Sigma_1^{-1}\Sigma_0) + (\mu_1 - \mu_0)^T \Sigma_1^{-1}(\mu_1 - \mu_0) - k + \log(\frac{det\,\Sigma_1}{det\,\Sigma_0}))$$

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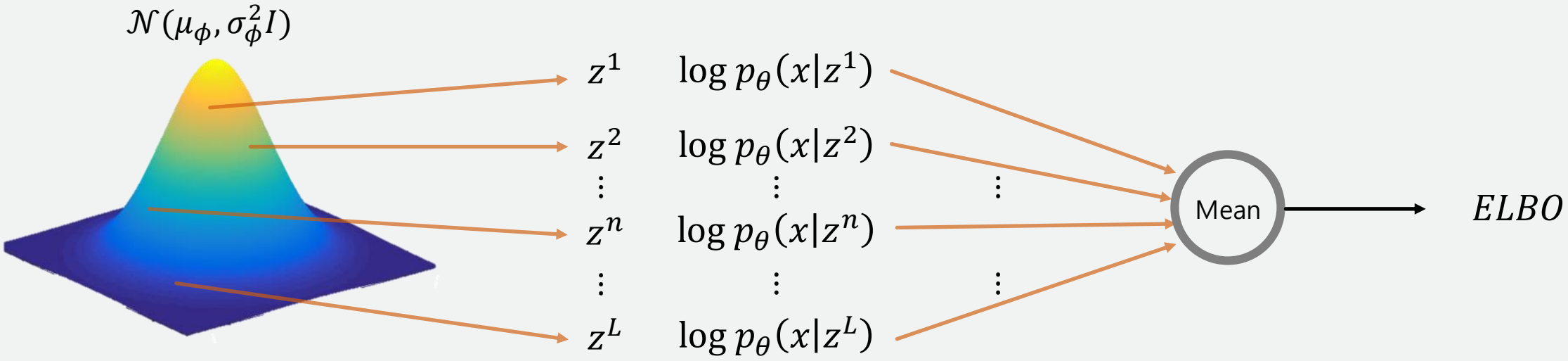
$$= \frac{1}{2}\left(\sum_j \sigma_{\phi_j}^2 + \sum_j \mu_{\phi_j}^2 - J - \sum_j \log \sigma_{\phi_j}^2\right)$$

$$= \frac{1}{2}\sum_j (\mu_{\phi_j}^2 + \sigma_{\phi_j}^2 - \log \sigma_{\phi_j}^2 - 1)$$

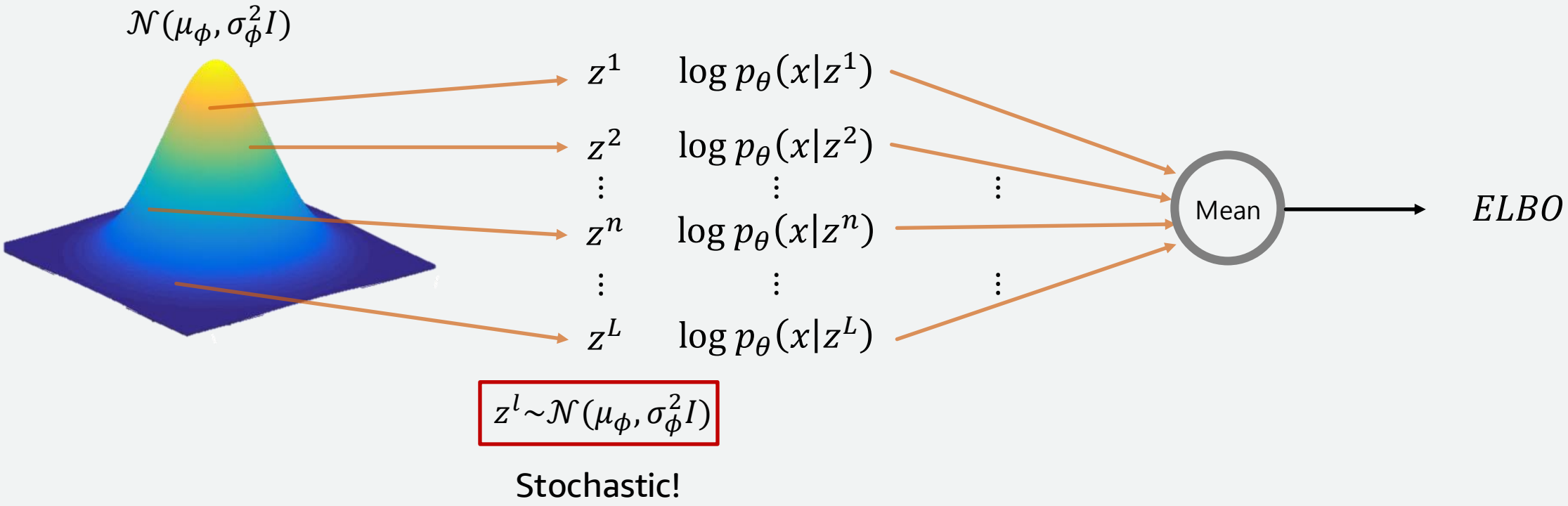
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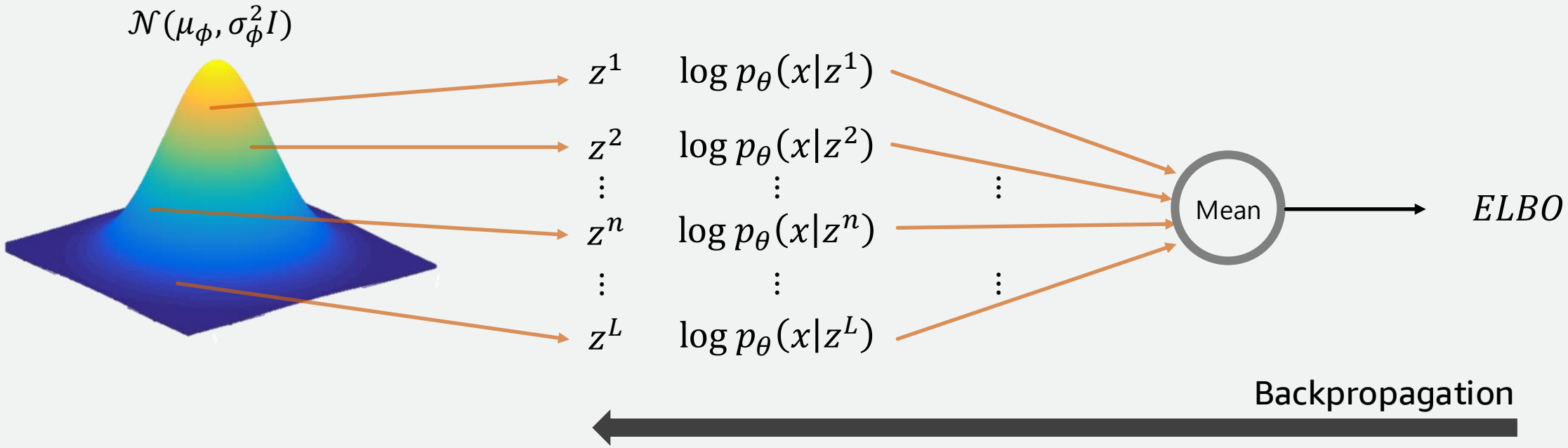
Variational Autoencoder(VAE)



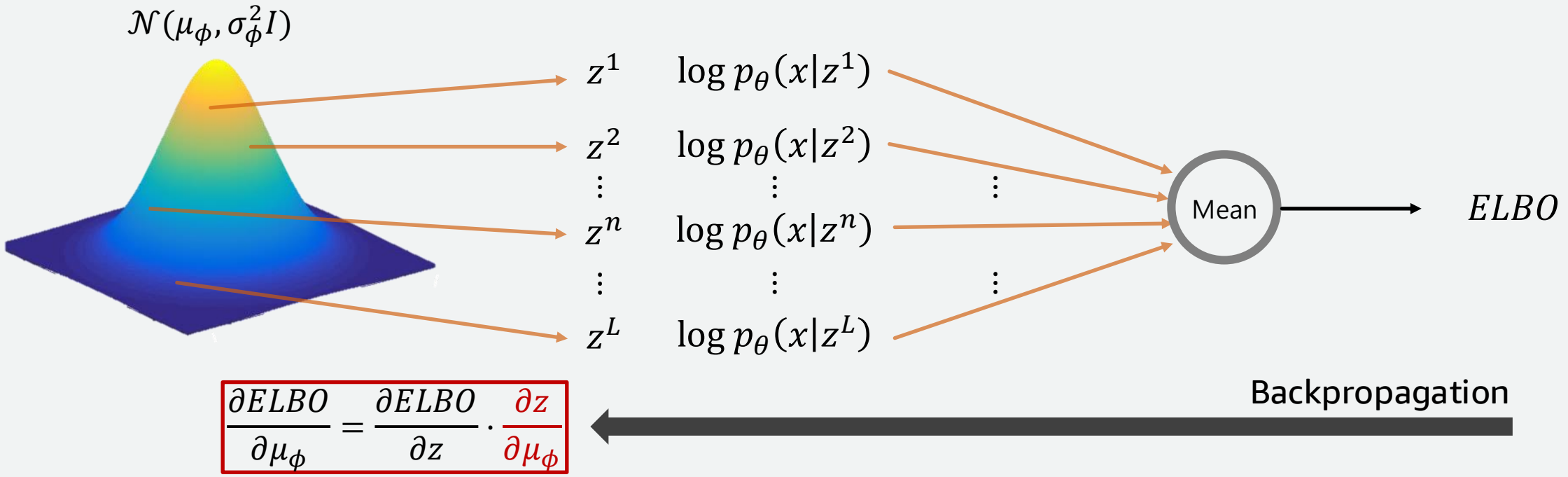
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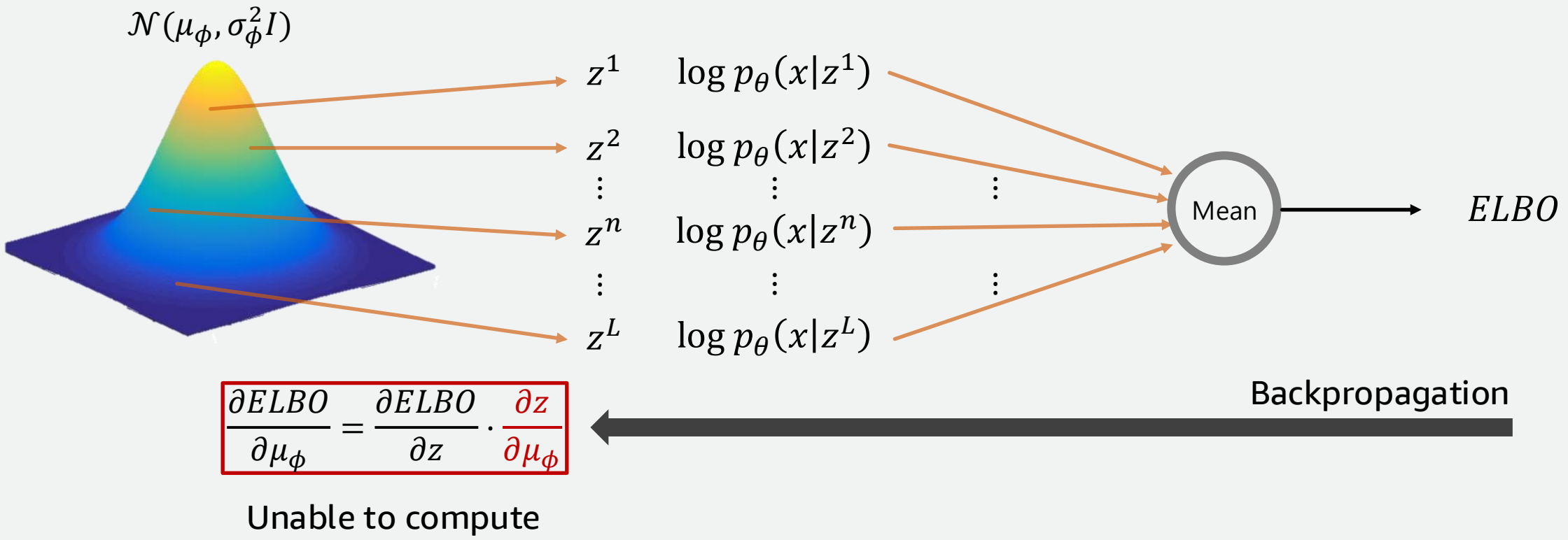
Variational Autoencoder(VAE)



Variational Autoencoder(VAE)



Variational Autoencoder(VAE)



REINFORCE: Log-derivative trick

$$\nabla_{\phi}\left(\mathbb{E}_{q_{\phi}}[\log p_{\theta}(x|z)] - D_{KL}\left(q_{\phi}(z|x)||p(z)\right)\right)$$

REINFORCE: Log-derivative trick

$$\nabla_{\phi}\left(\mathbb{E}_{q_{\phi}}[\log p_{\theta}(x|z)] - D_{KL}\left(q_{\phi}(z|x)||p(z)\right)\right)$$

$$= \nabla_{\phi}\left(\mathbb{E}_{q_{\phi}}[\log p_{\theta}(x|z)] - \mathbb{E}_{q_{\phi}}\left[\log q_{\phi}(z|x) - \log p(z)\right]\right) \quad \cdots \text{ (By definition)}$$

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$$= \int \nabla_{\phi}[(q_{\phi}(z|x)\left[\log p_{\theta}(x|z) + \log p(z) - \log q_{\phi}(z|x)\right]]dz \quad \cdots \text{ (By Expectation \& Leibniz Rule)}$$

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$$= \int \left(\nabla_{\phi} q_\phi(z|x)\right)\left[\log p_\theta(x|z) + \log p(z) - \log q_\phi(z|x)\right]dz \quad \cdots \text{ (By Product Rule)}$$

$$+\int q_\phi(z|x)\nabla_{\phi}\left[\log p_\theta(x|z) + \log p(z) - \log q_\phi(z|x)\right]dz$$

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$$\int \left(\nabla_{\phi} q_{\phi}(z|x)\right)\left[\log p_{\theta}(x|z) + \log p(z) - \log q_{\phi}(z|x)\right]dz$$

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Log-derivative trick!

REINFORCE: Log-derivative trick

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$$= \mathbb{E}_{q_\phi}\boxed{\nabla_\phi \log q_\phi(z|x)}\left[\log p_\theta(x|z) + \log p(z) - \log q_\phi(z|x)\right]$$

It's now differentiable since z fixed

REINFORCE: Log-derivative trick

$$\int \left(\nabla_{\phi} q_{\phi}(z|x)\right)\left[\log p_{\theta}(x|z) + \log p(z) - \log q_{\phi}(z|x)\right]dz$$

$$+\int q_{\phi}(z|x)\nabla_{\phi}\left[\log p_{\theta}(x|z) + \log p(z) - \log q_{\phi}(z|x)\right]dz$$

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$$= \mathbb{E}_{q_{\phi}}\nabla_{\phi}\log q_{\phi}(z|x)\left[\log p_{\theta}(x|z) + \log p(z) - \log q_{\phi}(z|x)\right]$$

$$\approx \frac{1}{L}\sum_{l=1}^{L}\nabla_{\phi}\log q_{\phi}\left(z^{(l)}|x\right)\left[\log p_{\theta}\left(x|z^{(l)}\right) + \log p\left(z^{(l)}\right) - \log q_{\phi}\left(z^{(l)}|x\right)\right]$$

REINFORCE: Log-derivative trick

$$\nabla_\phi \mathrm{ELBO} \approx \frac{1}{L}\sum_{l=1}^{L} \nabla_\phi \log q_\phi\left(z^{(l)}|x\right)\left[\log p_\theta\left(x|z^{(l)}\right) + \log p\left(z^{(l)}\right) - \log q_\phi\left(z^{(l)}|x\right)\right]$$

High variance

VAE Optimization – Reparameterization trick

Divide sampling in two step:

VAE Optimization – Reparameterization trick

Divide sampling in two step:

$$\epsilon \sim p'(\epsilon), where\ p'\ is\ simple\ distribution\big(e.g., \mathcal{N}(0, I)\big)$$

$$z = g_{\phi}(\epsilon, x)$$

VAE Optimization – Reparameterization trick

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If $q_\phi(z|x)\ Gaussian \rightarrow z = g_\phi(\epsilon, x) = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$

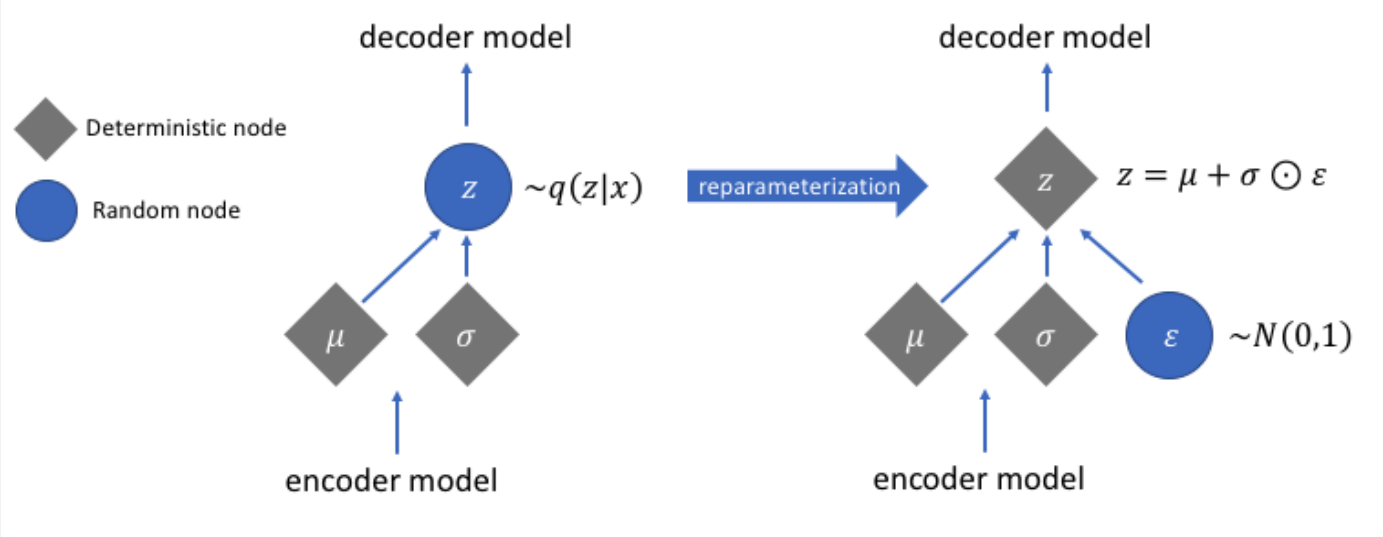
VAE Optimization – Reparameterization trick

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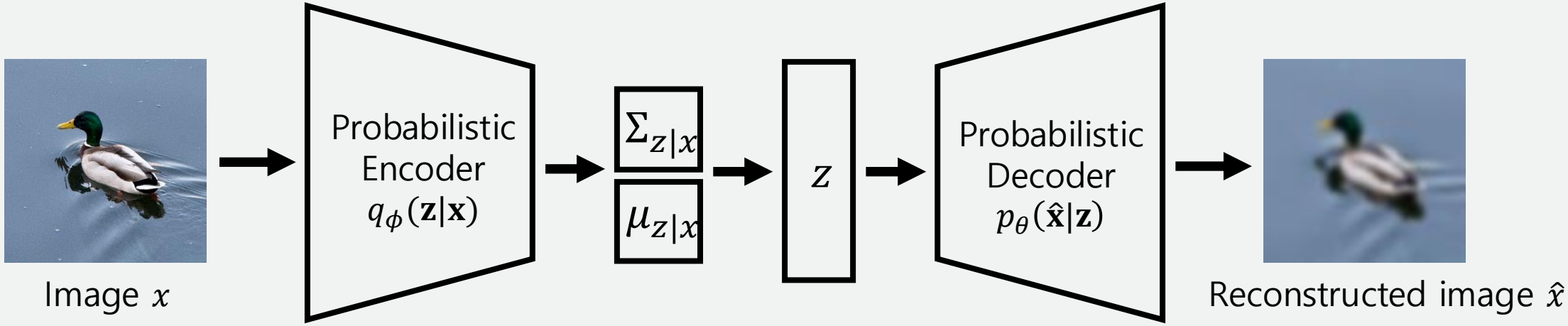
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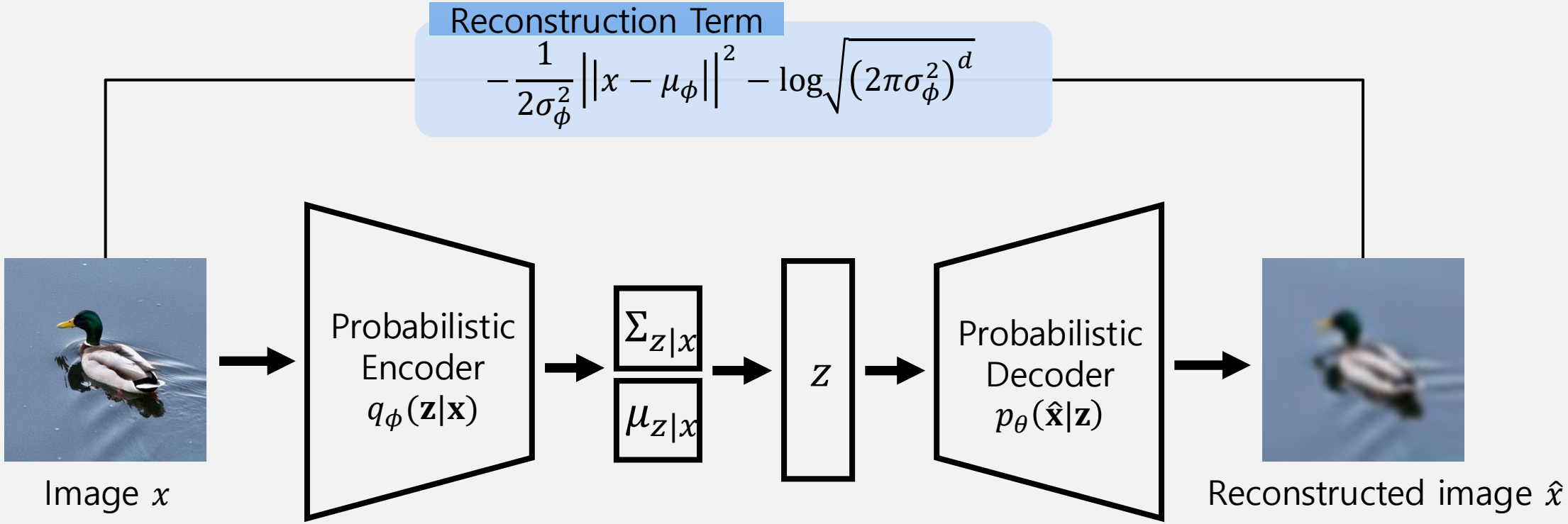
If $q_\phi(z|x)\ Gaussian \rightarrow z = g_\phi(\epsilon, x) = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$



VAE Optimization – Summary



VAE Optimization – Summary



VAE Optimization – Summary

Reconstruction Term

$$-\frac{1}{2\sigma_\phi^2}\left|\left|x-\mu_\phi\right|\right|^2-\log\sqrt{\left(2\pi\sigma_\phi^2\right)^d}$$

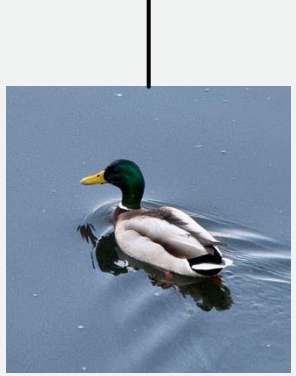


Image x

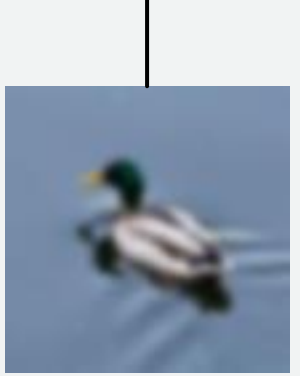
Probabilistic Encoder $q_\phi(\mathbf{z}|\mathbf{x})$

$\Sigma_{z|x}$

$\mu_{z|x}$

z

Probabilistic Decoder $p_\theta(\hat{\mathbf{x}}|\mathbf{z})$

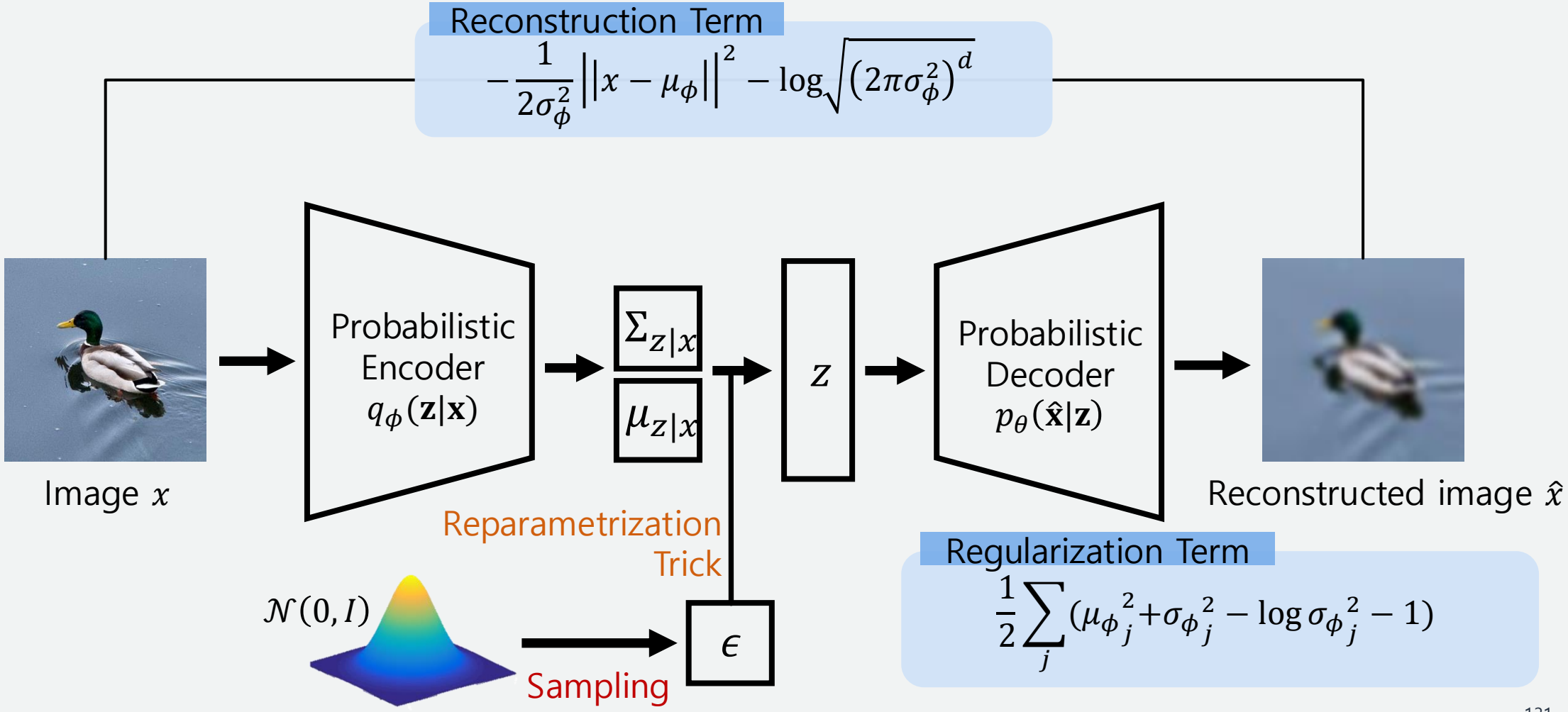


Reconstructed image $\hat{x}$

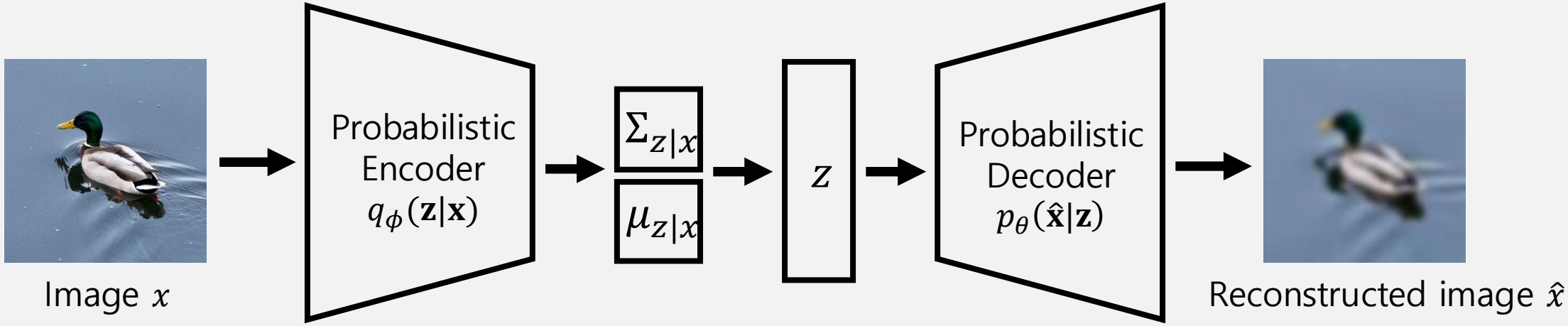
Regularization Term

$$\frac{1}{2}\sum_j\left(\mu_{\phi_j}^2+\sigma_{\phi_j}^2-\log\sigma_{\phi_j}^2-1\right)$$

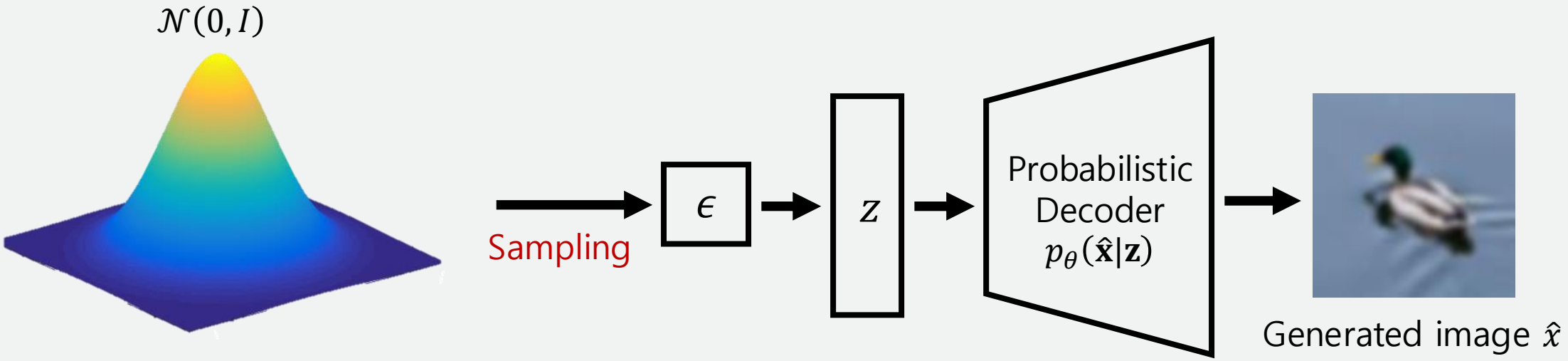
VAE Optimization – Summary



VAE Generation



VAE Generation



DDPM

DDPM(Denoising Diffusion Probabilistic Models)

VAE: oversimplified latent prior, posterior collapse

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VAE: oversimplified latent prior, posterior collapse

$$ELBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

z is too simple to encoding real data

DDPM(Denoising Diffusion Probabilistic Models)

VAE: oversimplified latent prior, posterior collapse

$$\mathrm{E}LBO = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x) \,||\, p(z)\right)$$

Powerful decoder encourages the encoder to match the prior

DDPM(Denoising Diffusion Probabilistic Models)

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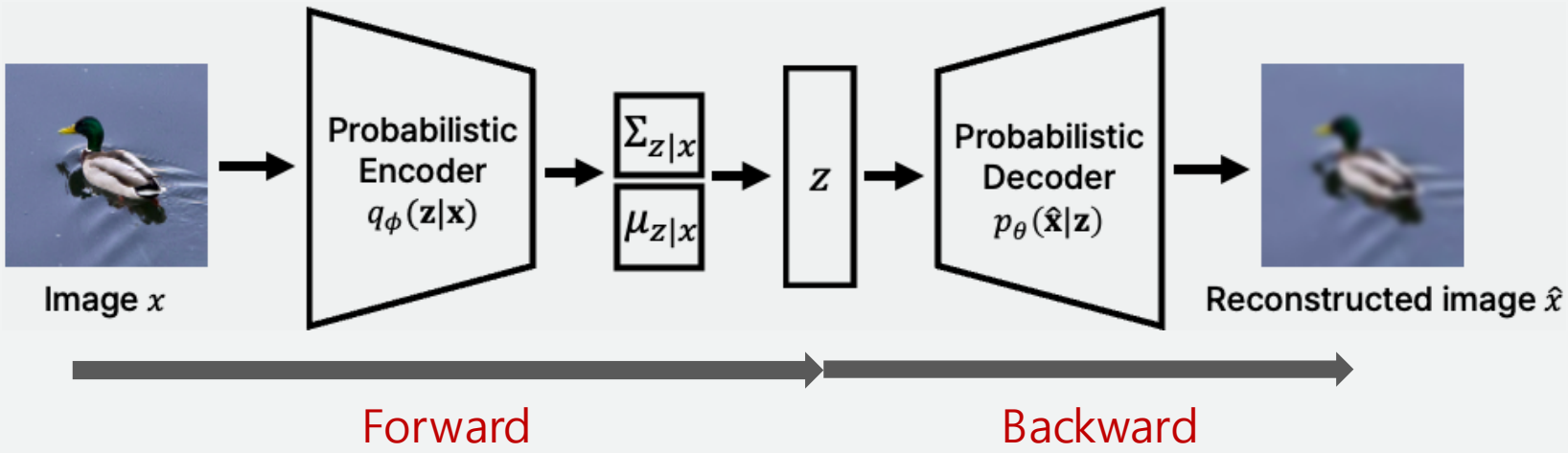
Useful resource: VQ-VAE

DDPM(Denoising Diffusion Probabilistic Models)

Diffusion models are latent variable models.

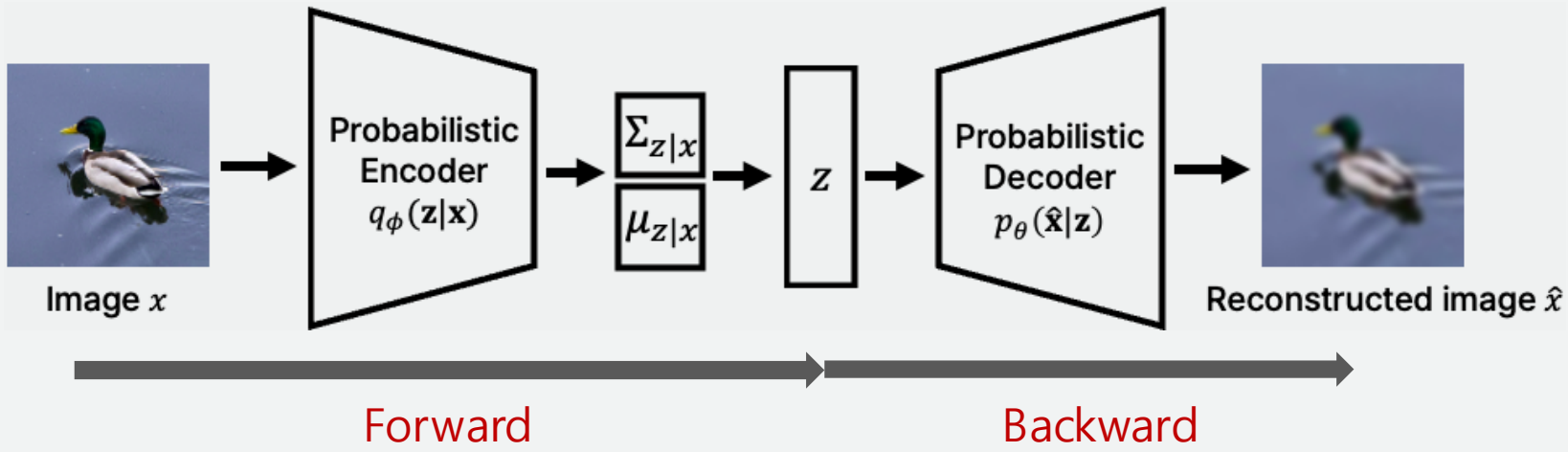
DDPM(Denoising Diffusion Probabilistic Models)

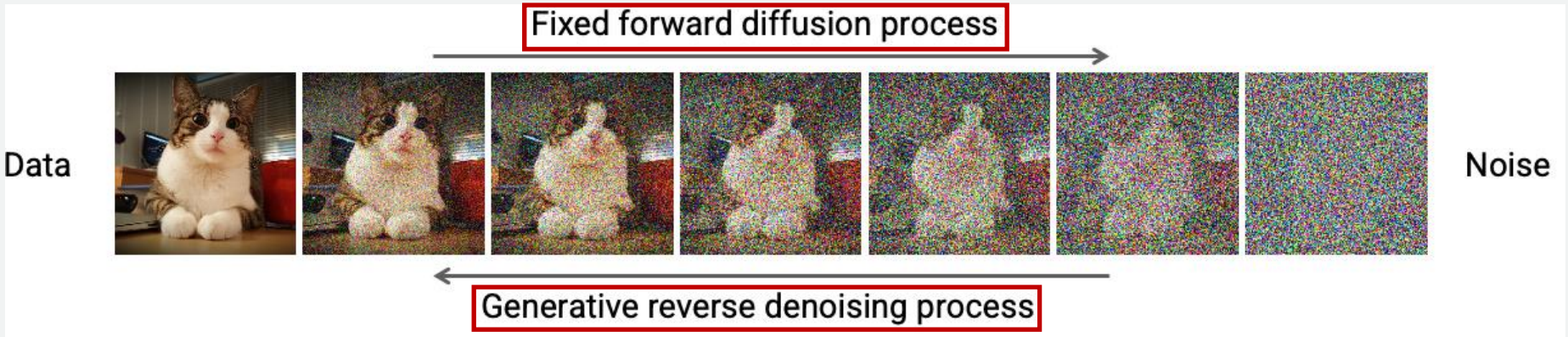
Diffusion models are latent variable models.



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DDPM: Maximizes ELBO with $q(x_t|x_0)$, $p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$

DDPM(Denoising Diffusion Probabilistic Models)

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Latent

Latent

DDPM(Denoising Diffusion Probabilistic Models)

Diffusion models are latent variable models.

VAE: Maximizes ELBO with $q_\phi(z|x)$, $p_\theta(x|z)$

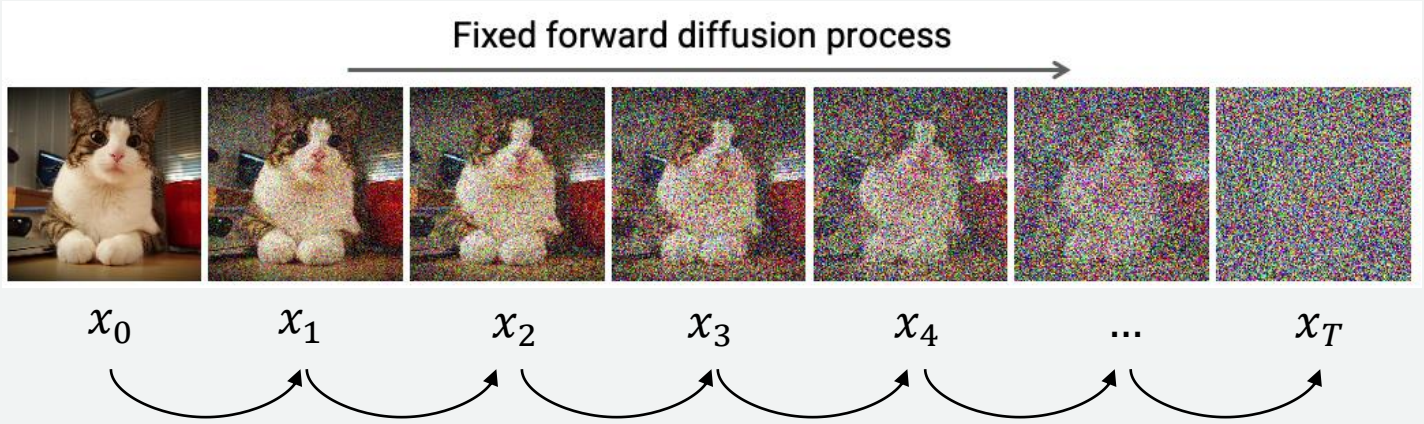
DDPM: Maximizes ELBO with $q(x_t|x_0)$, $p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$



Data(Image)

DDPM(Denoising Diffusion Probabilistic Models)

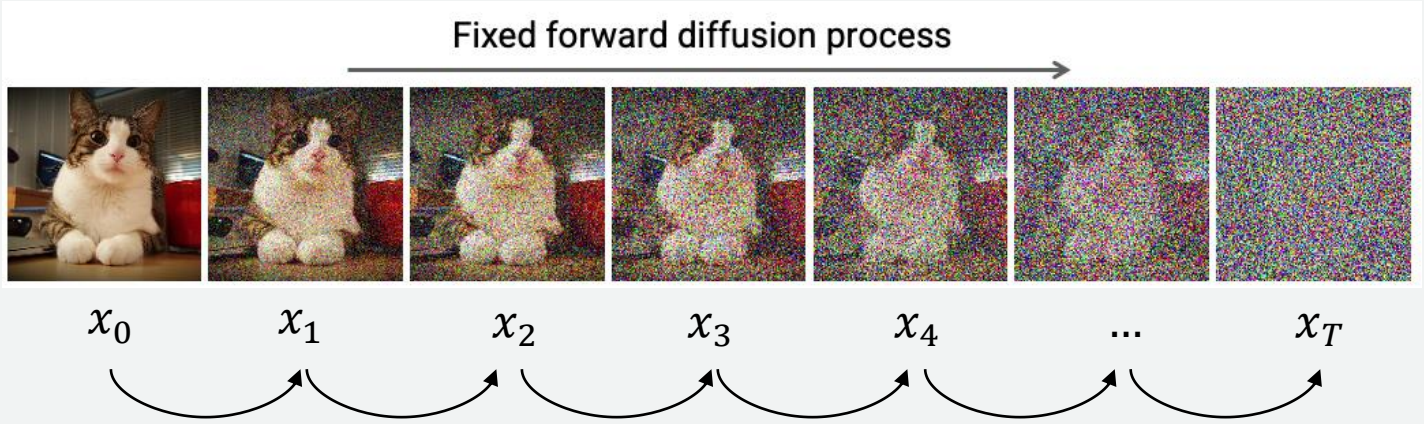
Assumes diffusion process as Markov chain that gradually adds Gaussian noise.



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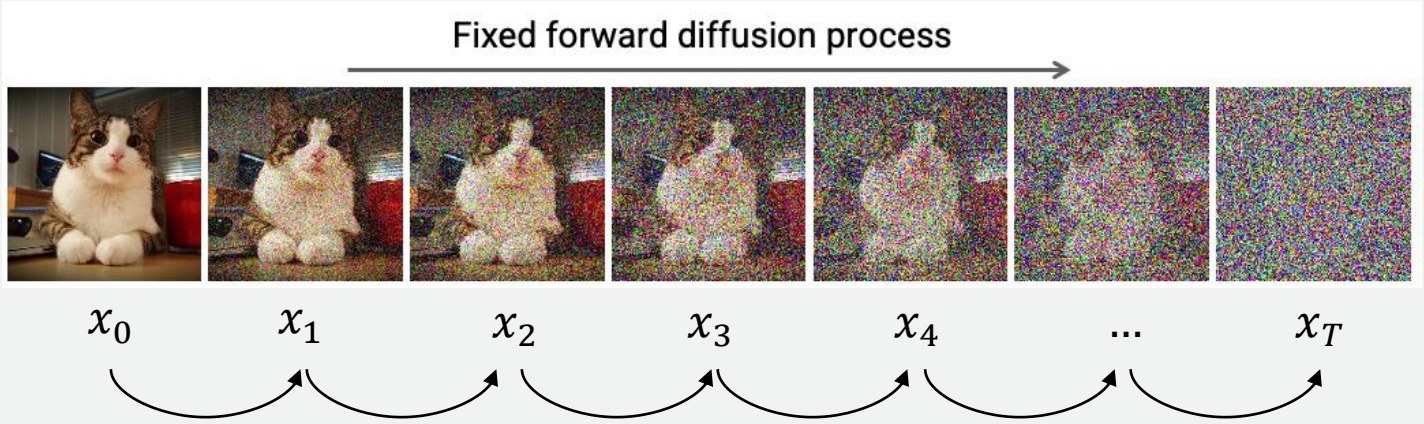


$$q(x_t|x_{t-1}, x_{t-2} \cdots x_0) = q(x_t|x_{t-1}) \coloneqq \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

Variance schedule

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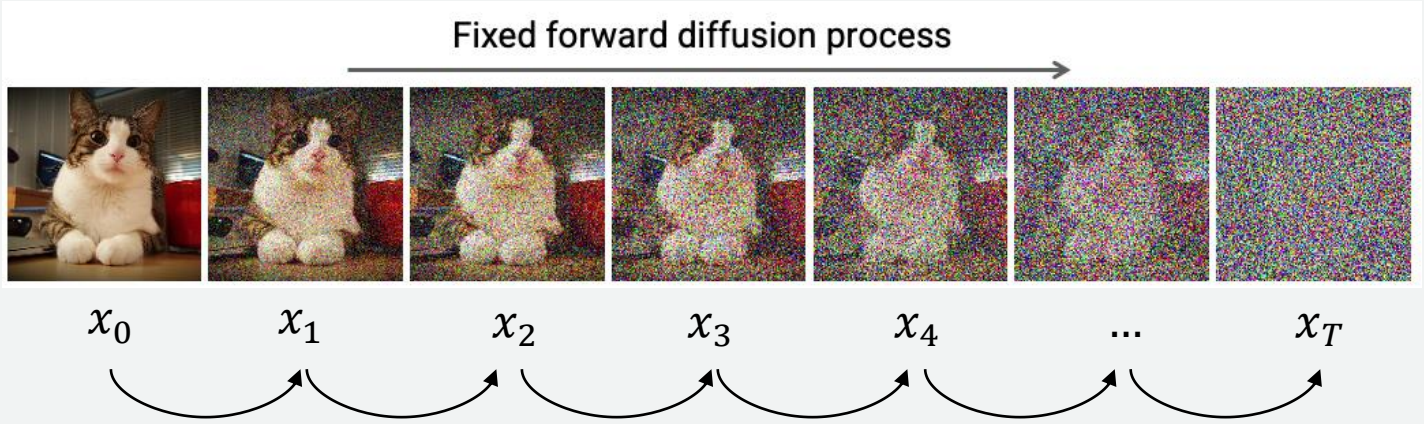


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DDPM(Denoising Diffusion Probabilistic Models)

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We will revisit this in <Diffusion as SDE(Wiener Process / Brownian Motion)>..

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How can we optimize this?

Next seminar(in two weeks)

- DDPM
  - Mathematical foundations
  - Training objectives
  - Sample generation
- DDPM as an SDE
- Advanced methods(DDIM, DPM-solver..)

Thank you!